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Im et al.

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(54) **REFLECTING MODULE FOR OPTICAL
IMAGE STABILIZATION (OIS) AND
CAMERA MODULE INCLUDING THE SAME**

(52) **U.S. Cl.**
CPC **G02B 27/646** (2013.01); **G02B 7/021**
(2013.01); **G02B 7/08** (2013.01); **G02B 7/09**
(2013.01);

(Continued)

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 542 days.

This patent is subject to a terminal dis-
claimer.

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lish and 6 pages in Korean).

(Continued)

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Apr. 29, 2020, now Pat. No. 11,262,593, which is a
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(74) *Attorney, Agent, or Firm* — NSIP Law

(30) **Foreign Application Priority Data**

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Apr. 17, 2017 (KR) 10-2017-0049048

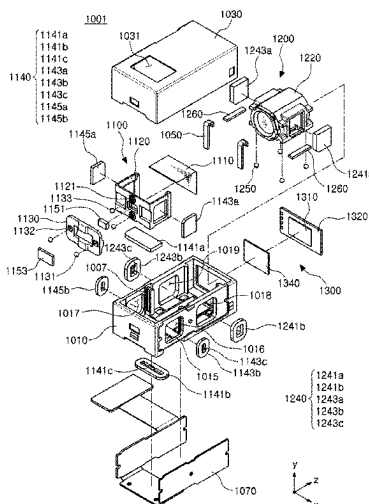
(57) **ABSTRACT**

A reflecting module for optical image stabilization (OIS)
includes a housing; a rotation holder provided in the housing
and comprising a reflecting member; a rotation plate pro-
vided in the housing between an inner wall of the housing
and the rotation holder so that the rotation holder is sup-
ported by the inner wall of the housing via the rotation plate;
and a driving part configured to apply a driving force to the
rotation holder to move the rotation holder.

(51) **Int. Cl.**
G02B 27/64 (2006.01)
G02B 7/02 (2021.01)

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26 Claims, 28 Drawing Sheets



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G02B 7/09 (2021.01)
G02B 13/00 (2006.01)
G02B 26/08 (2006.01)
G03B 5/00 (2021.01)
H04N 23/68 (2023.01)
G03B 13/36 (2021.01)
G03B 17/17 (2021.01)

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CPC **G02B 13/0065** (2013.01); **G02B 26/0816**
(2013.01); **G02B 27/642** (2013.01); **G03B**
5/00 (2013.01); **H04N 23/68** (2023.01); **G03B**
13/36 (2013.01); **G03B 17/17** (2013.01); **G03B**
2205/0023 (2013.01); **G03B 2205/003**
(2013.01); **G03B 2205/0069** (2013.01)

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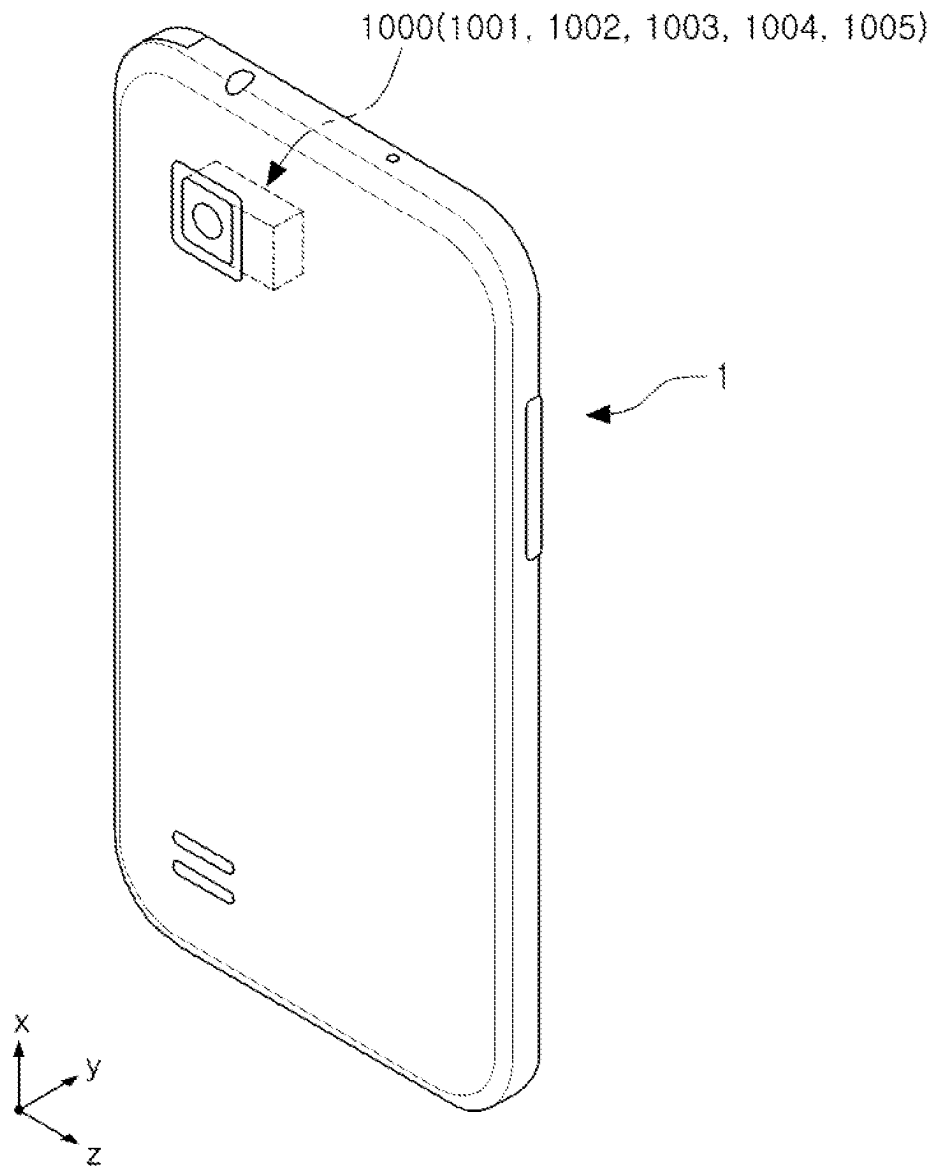


FIG. 1

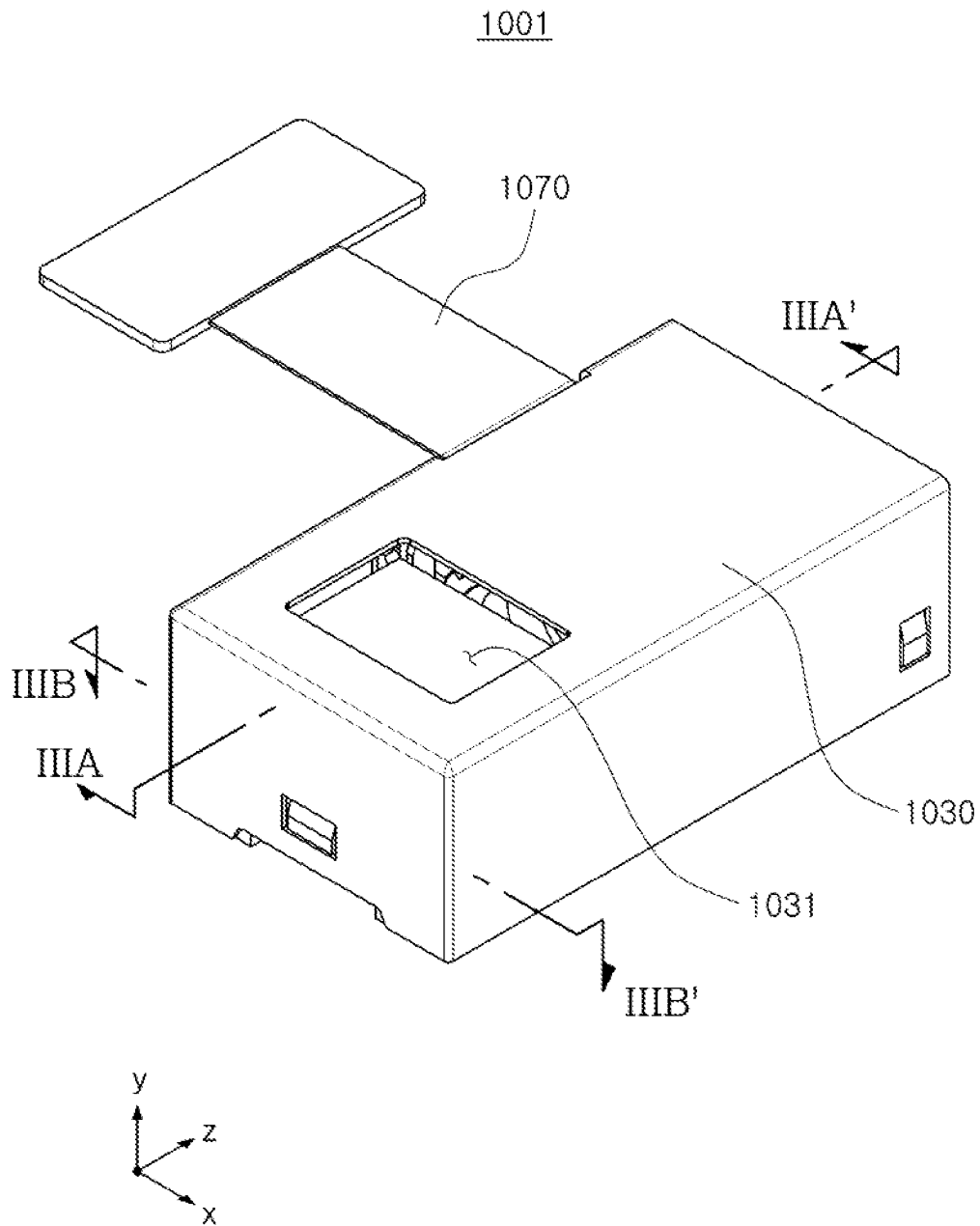


FIG. 2

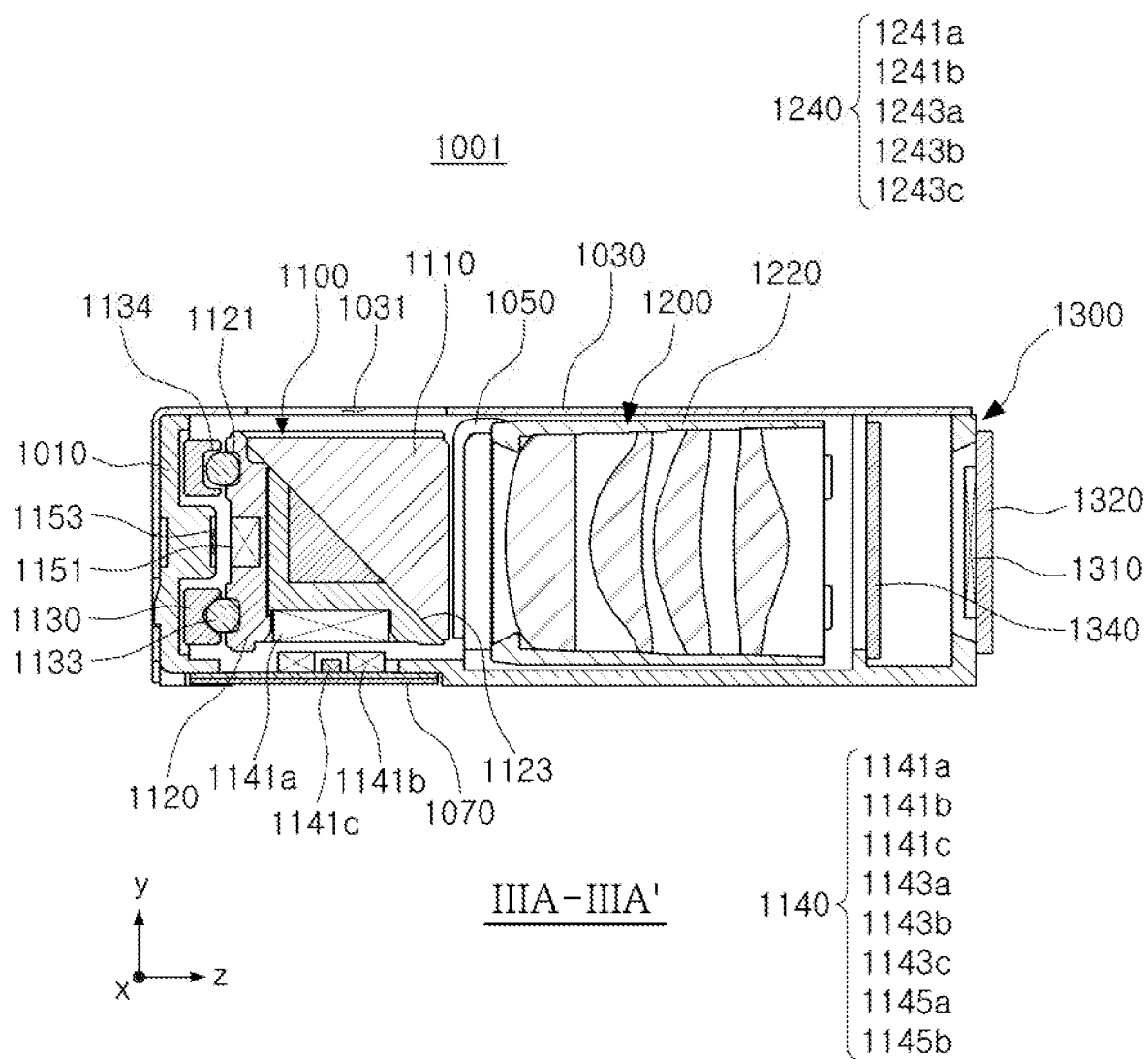


FIG. 3A

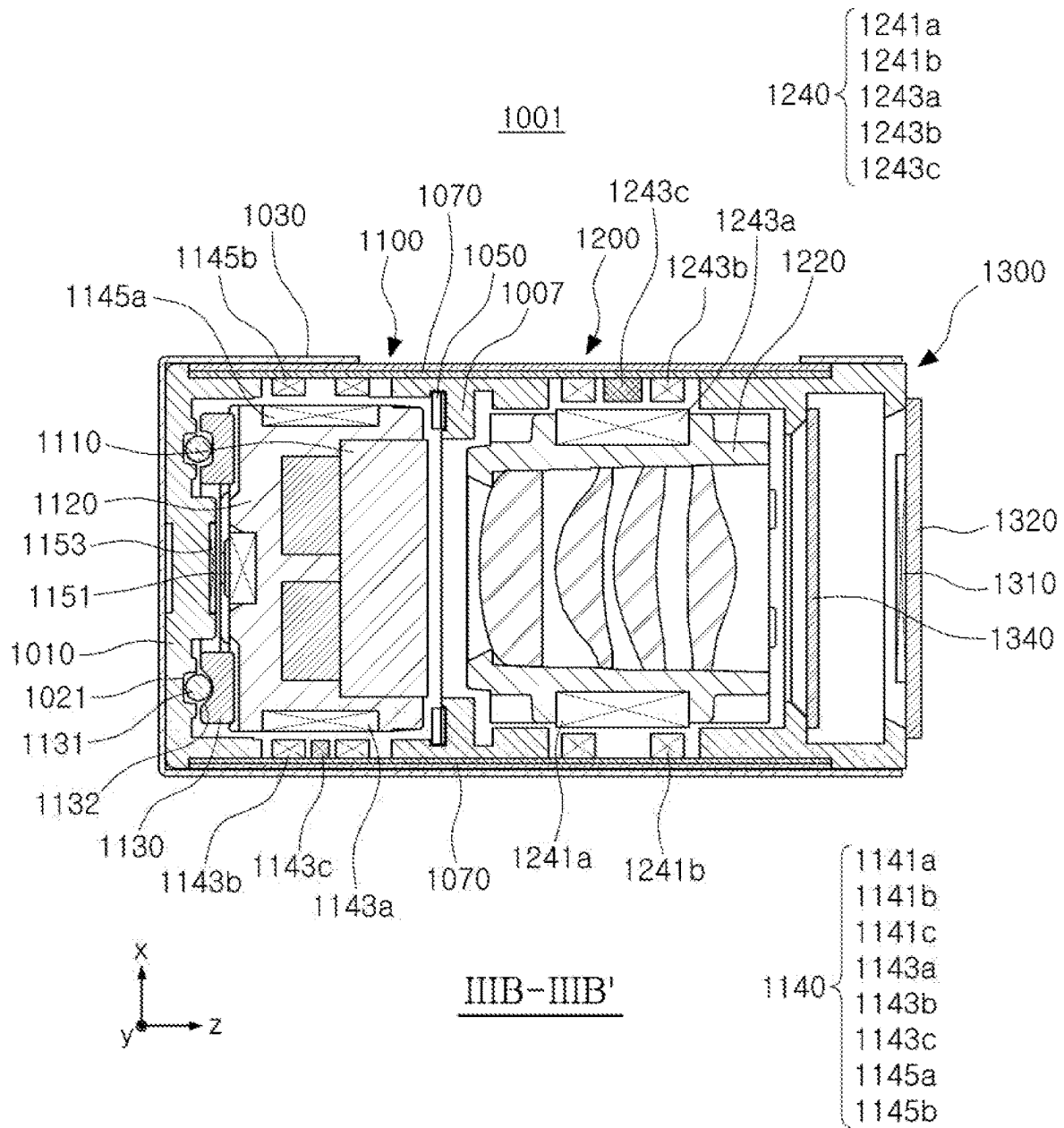


FIG. 3B

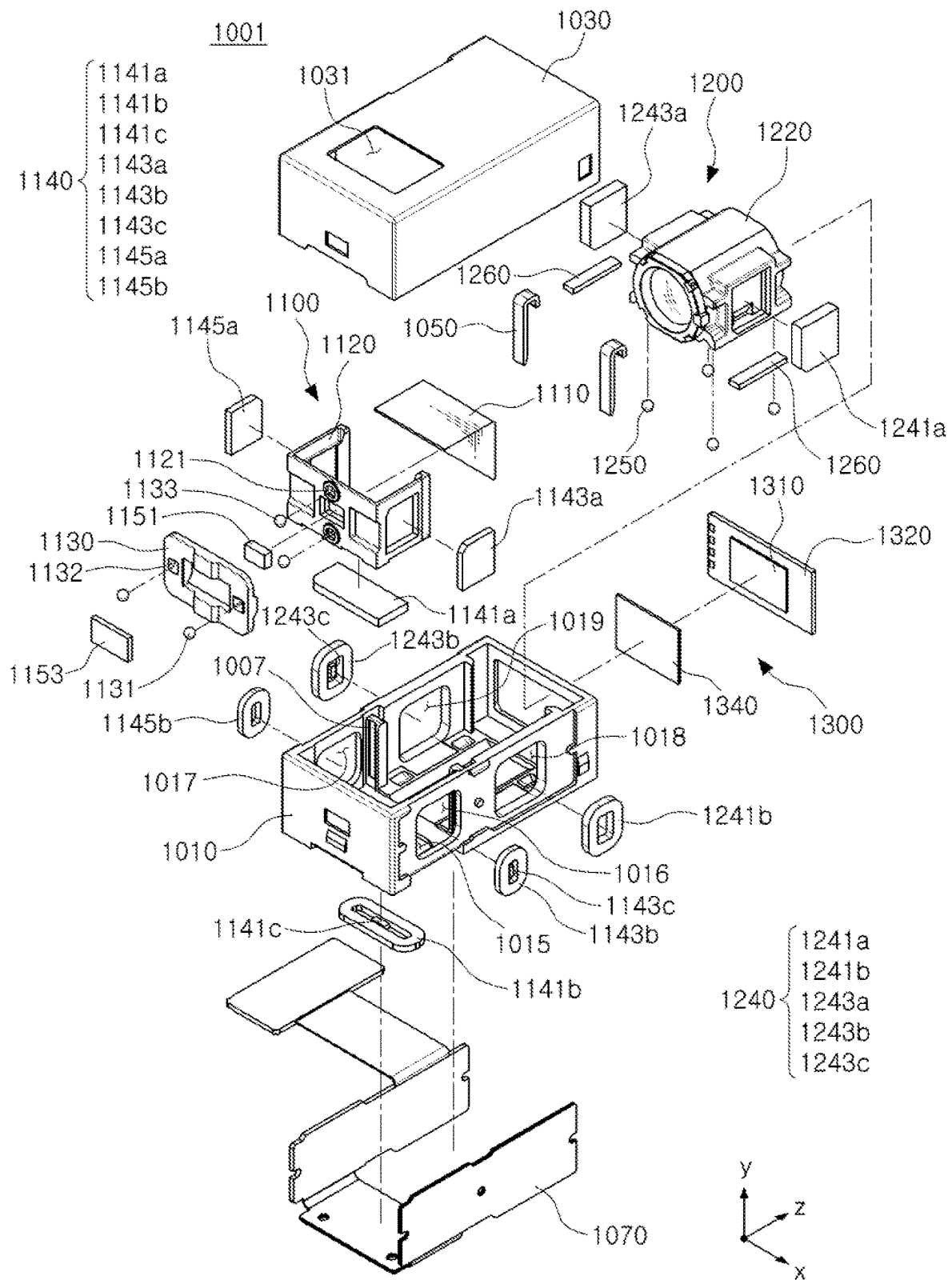


FIG. 4

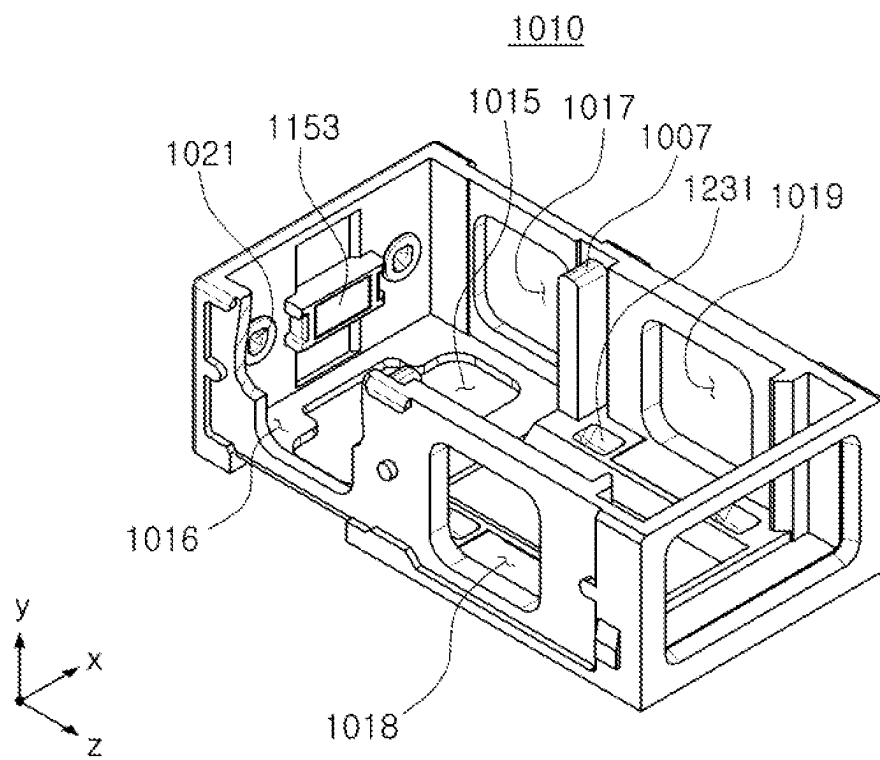


FIG. 5

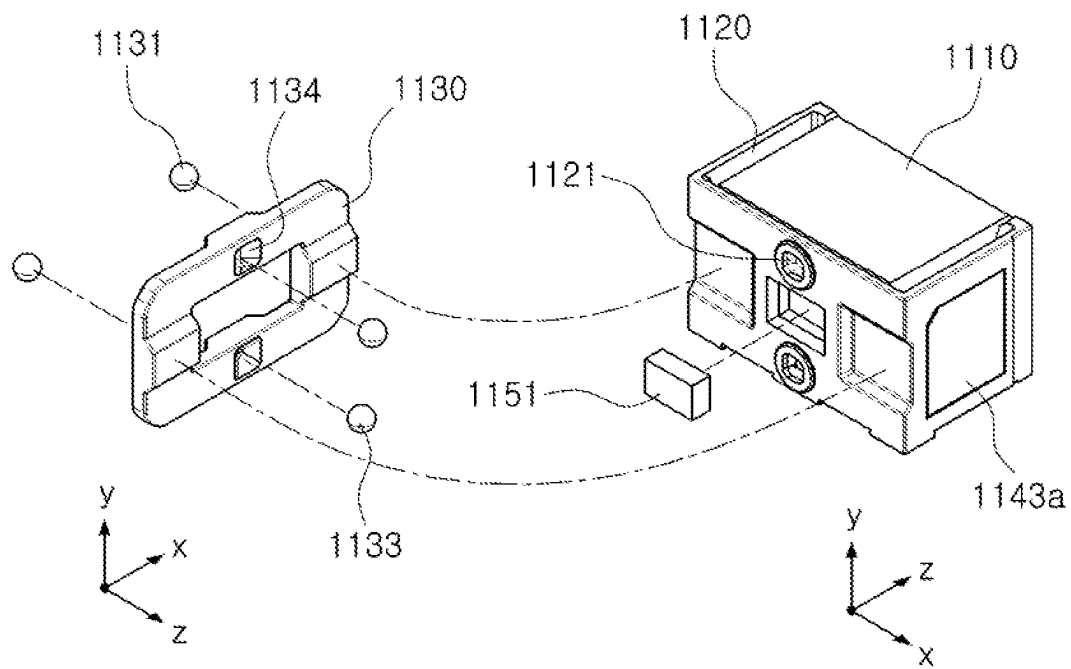


FIG. 6A

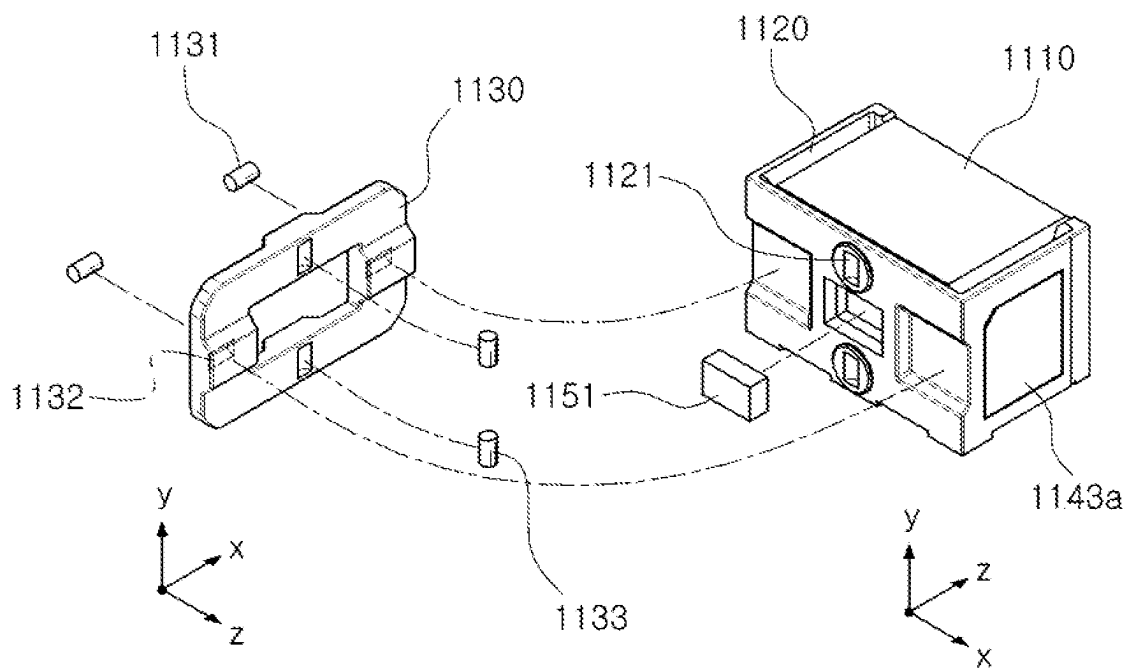
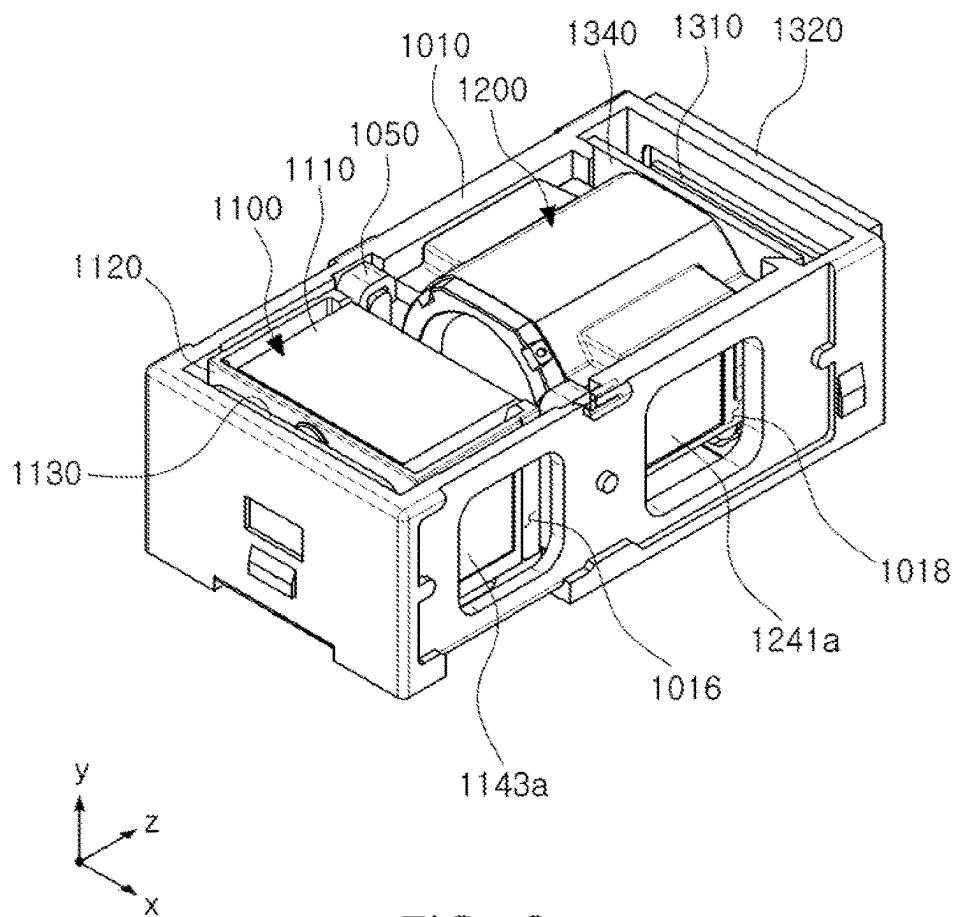
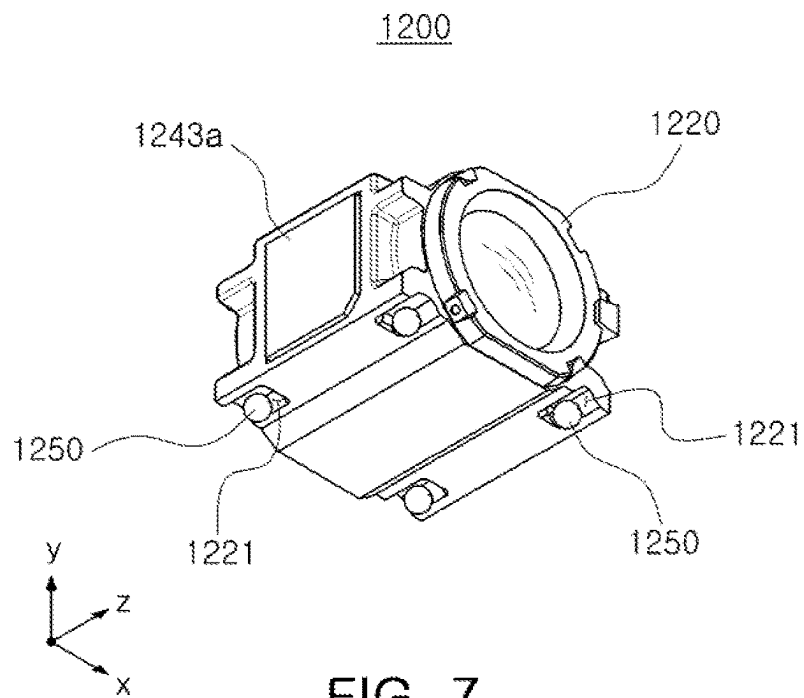


FIG. 6B



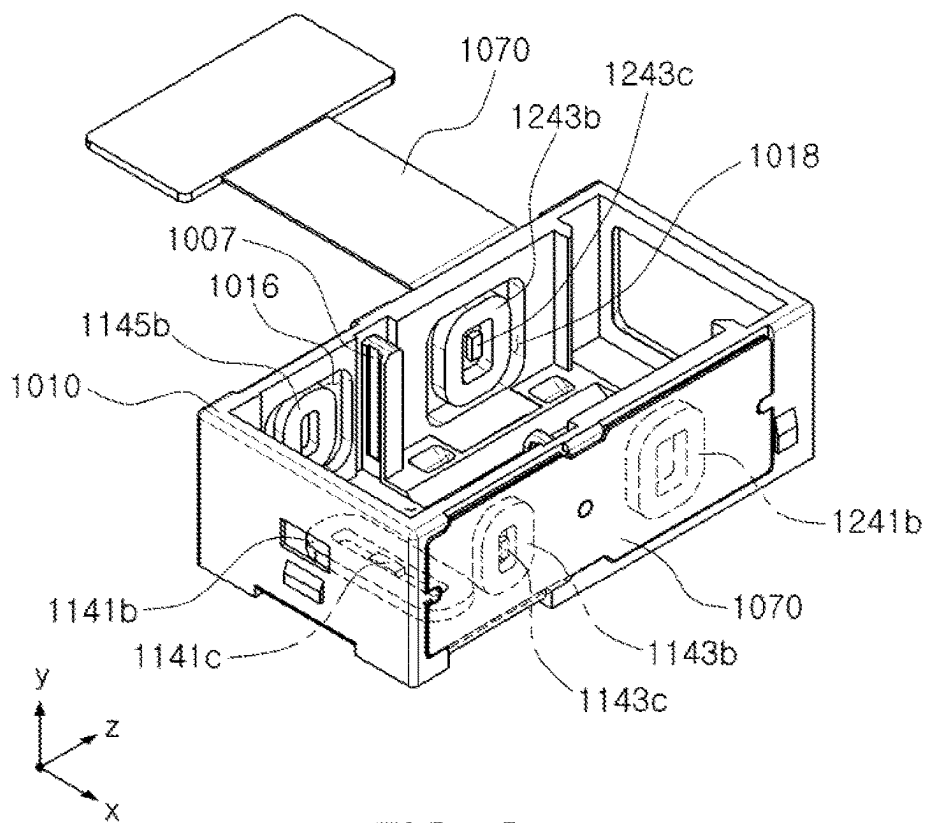


FIG. 9

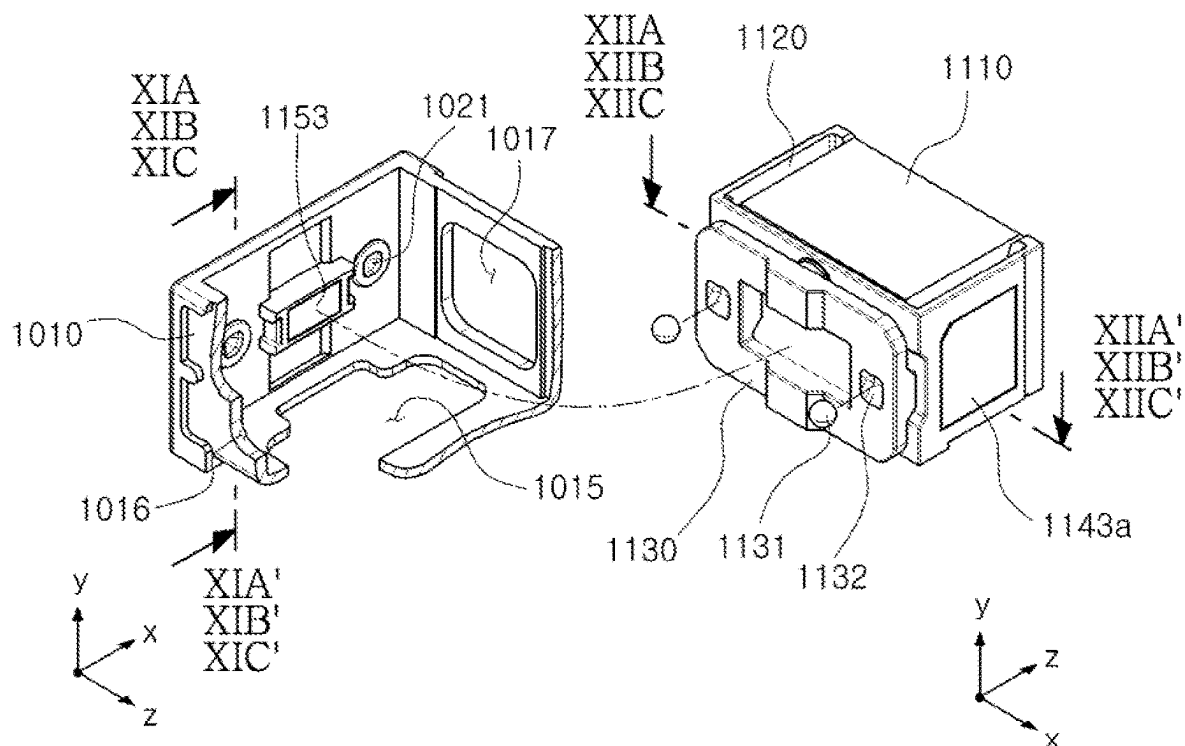


FIG. 10

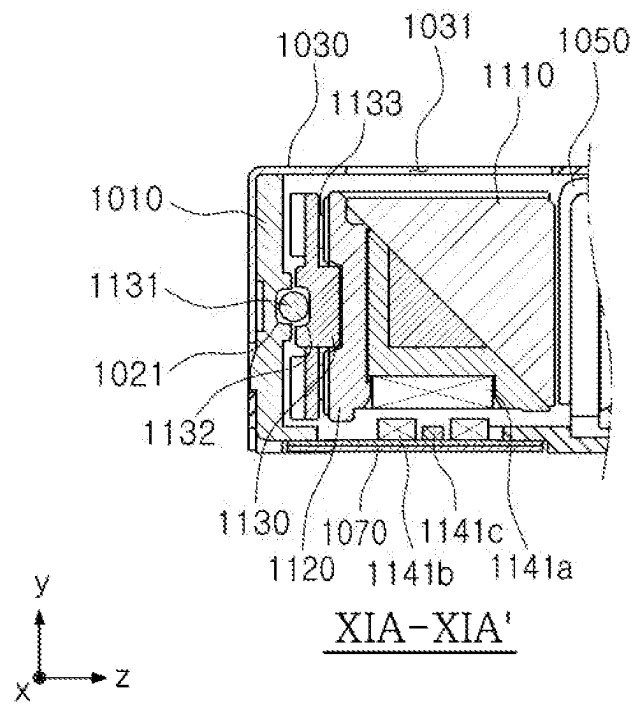


FIG. 11A

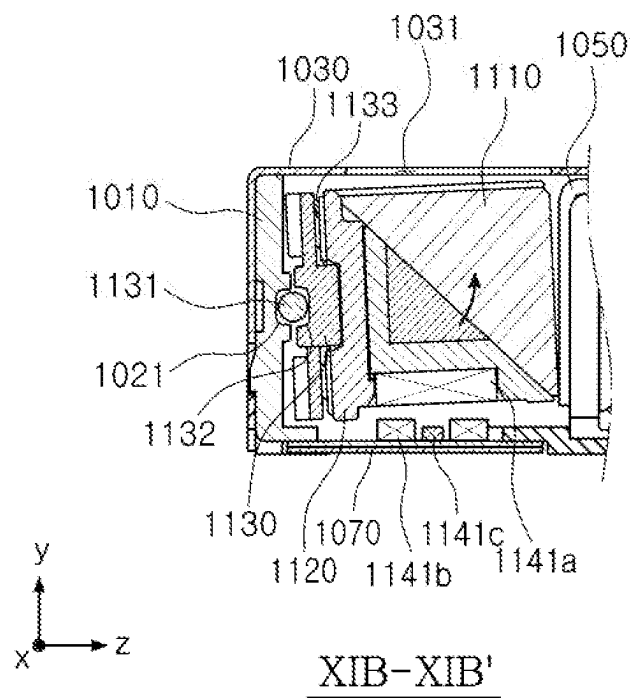


FIG. 11B

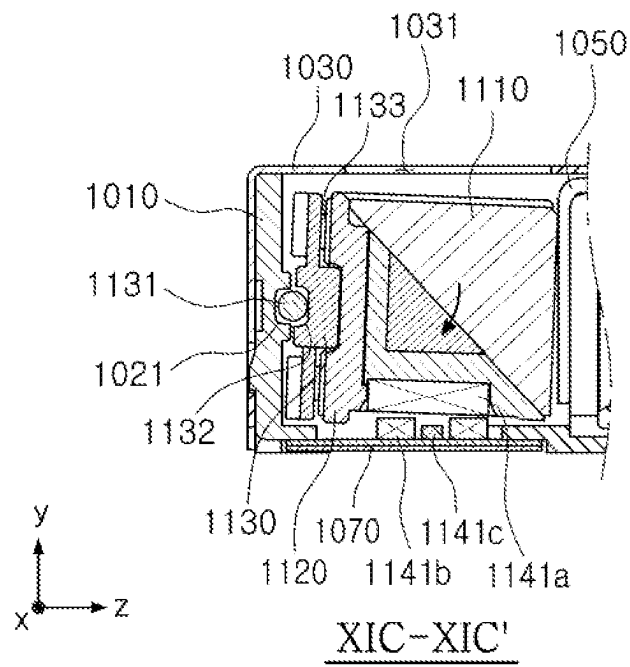


FIG. 11C

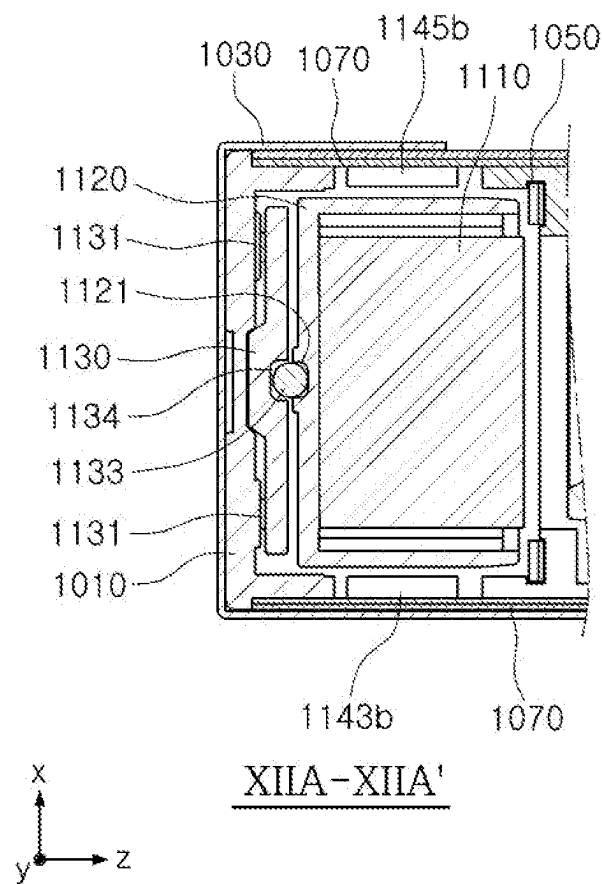
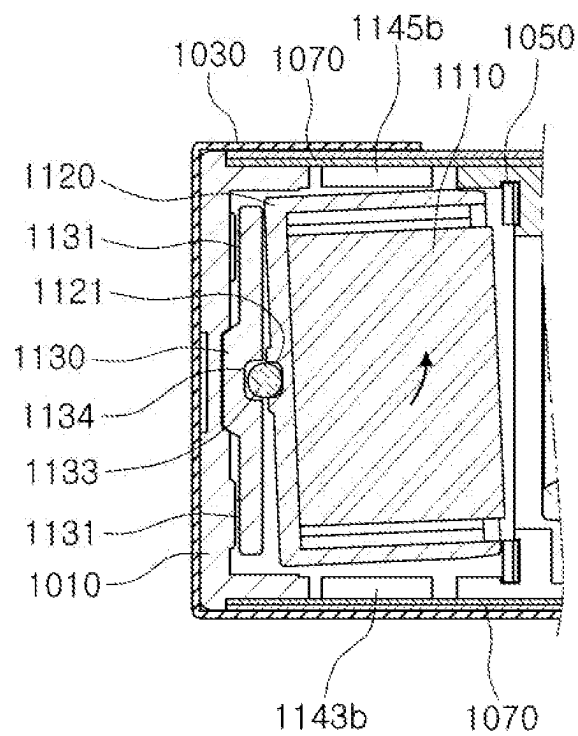


FIG. 12A



XIIB-XIIB'

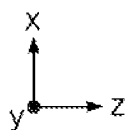
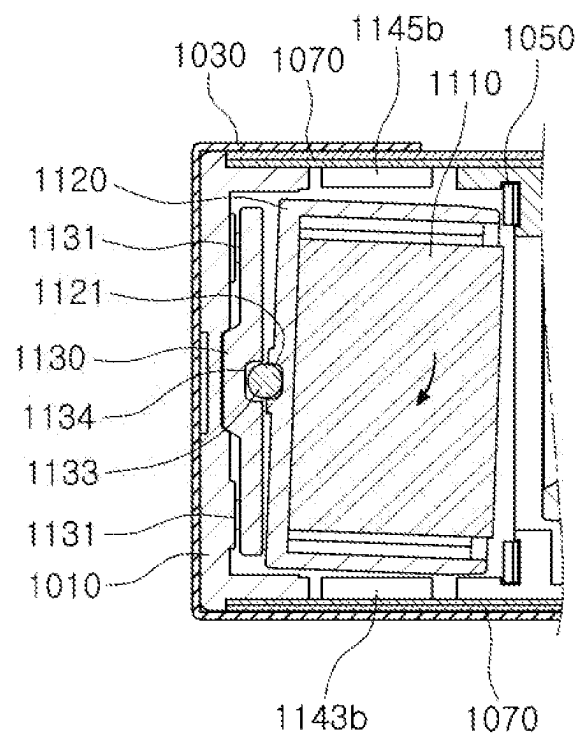


FIG. 12B



XIIC-XIIC'

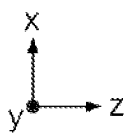


FIG. 12C

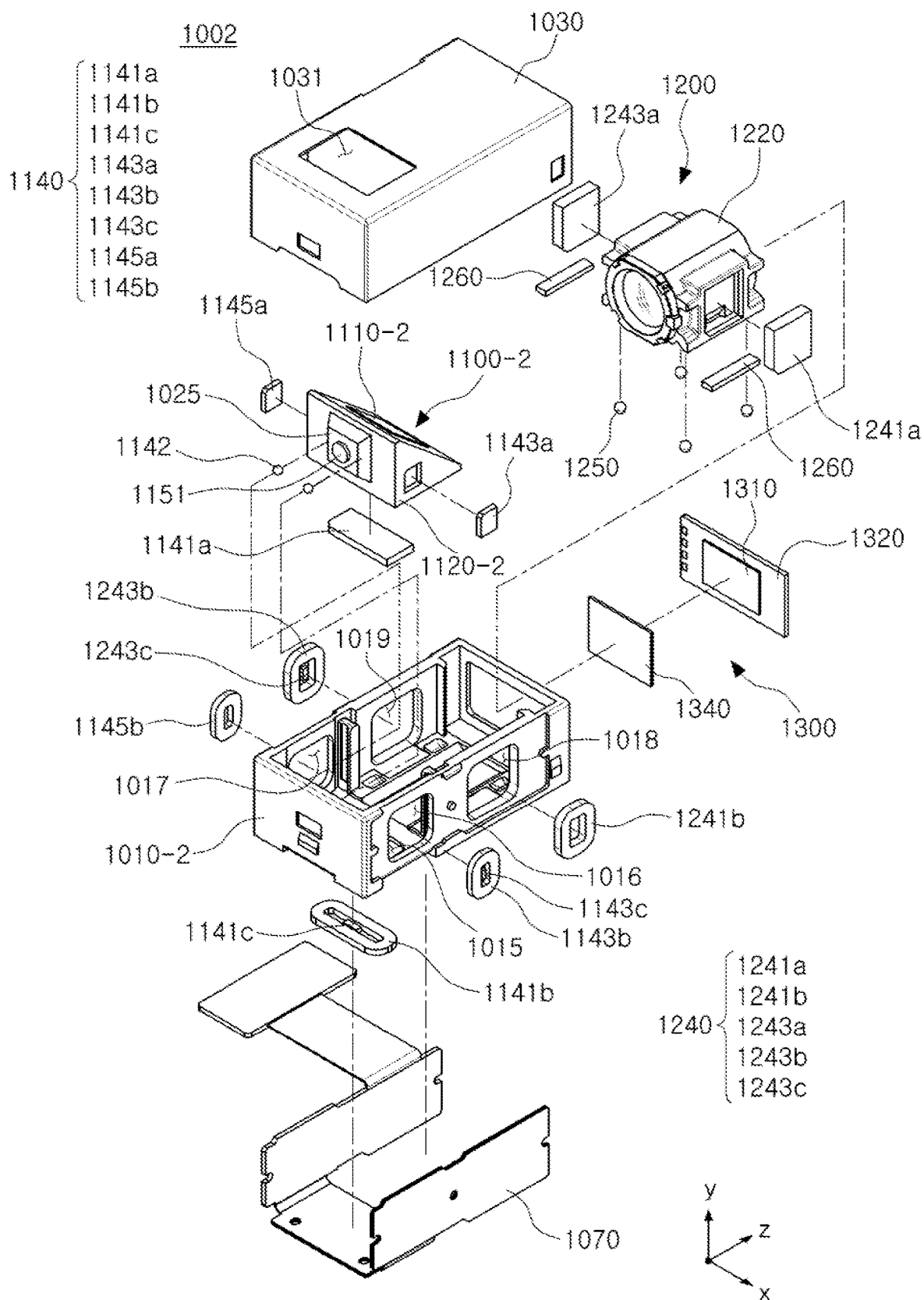


FIG. 13

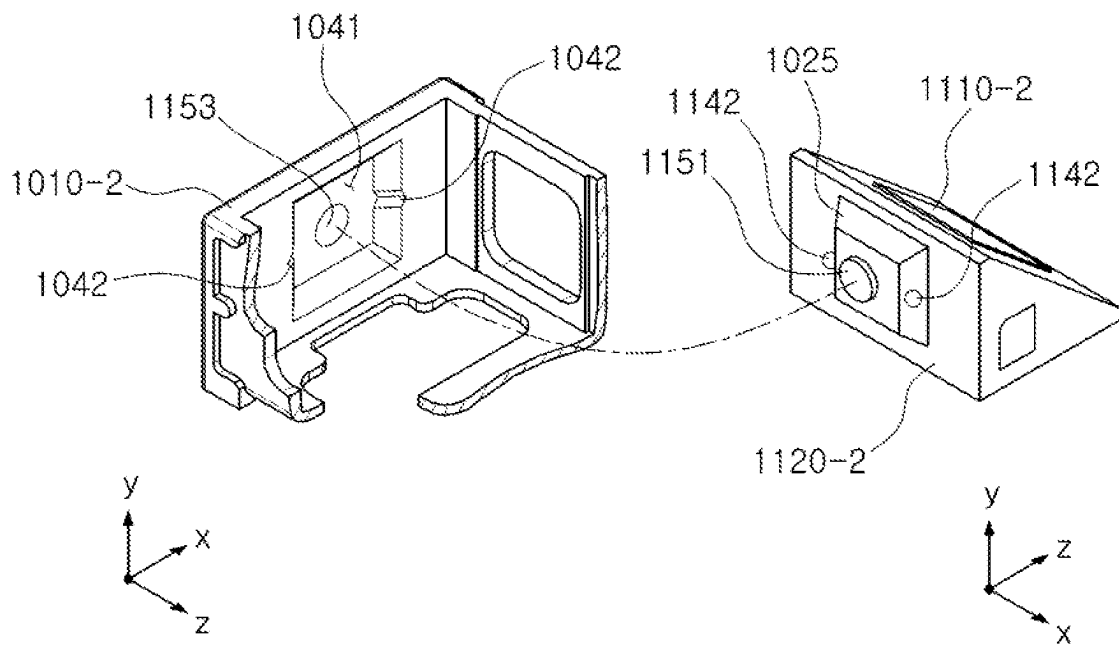


FIG. 14

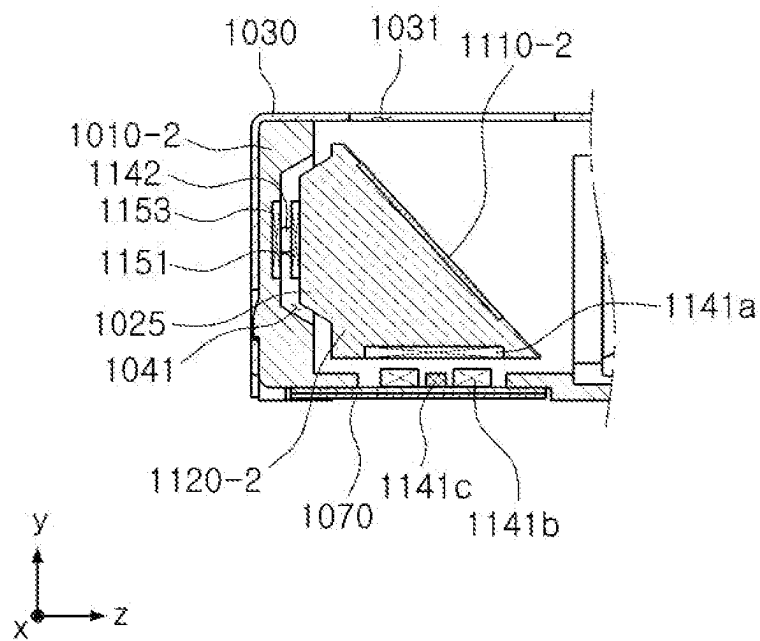


FIG. 15A

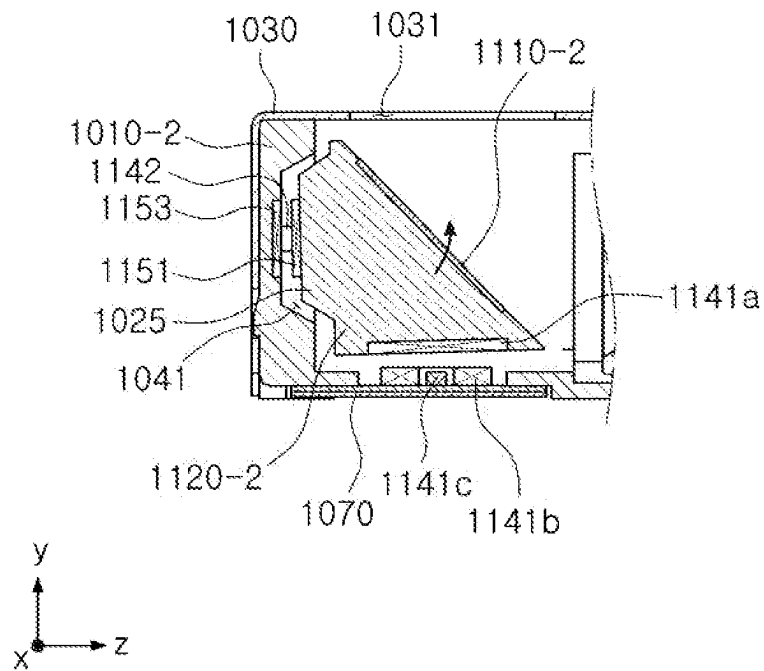


FIG. 15B

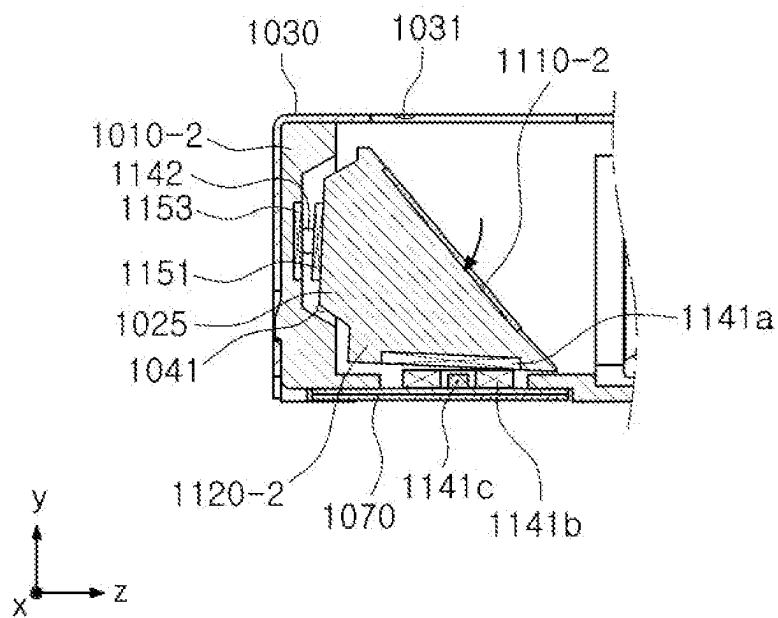
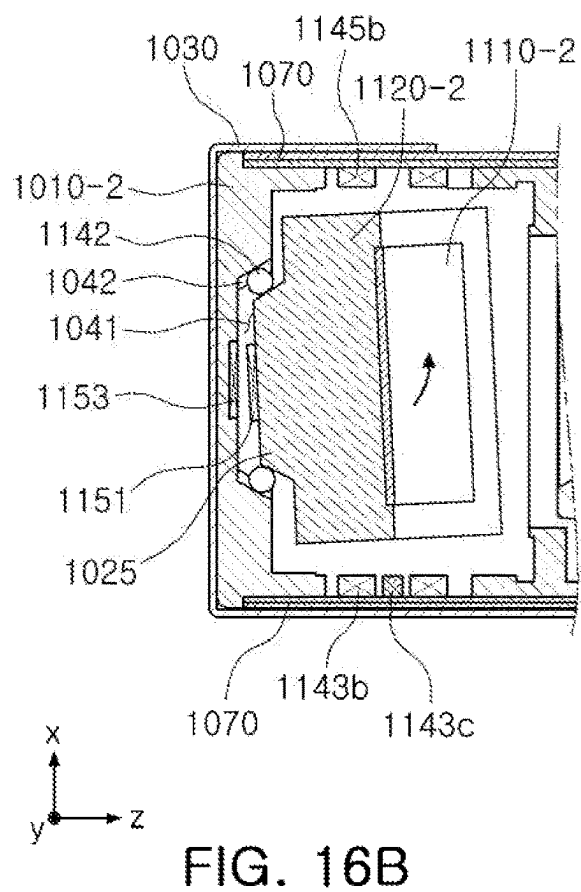
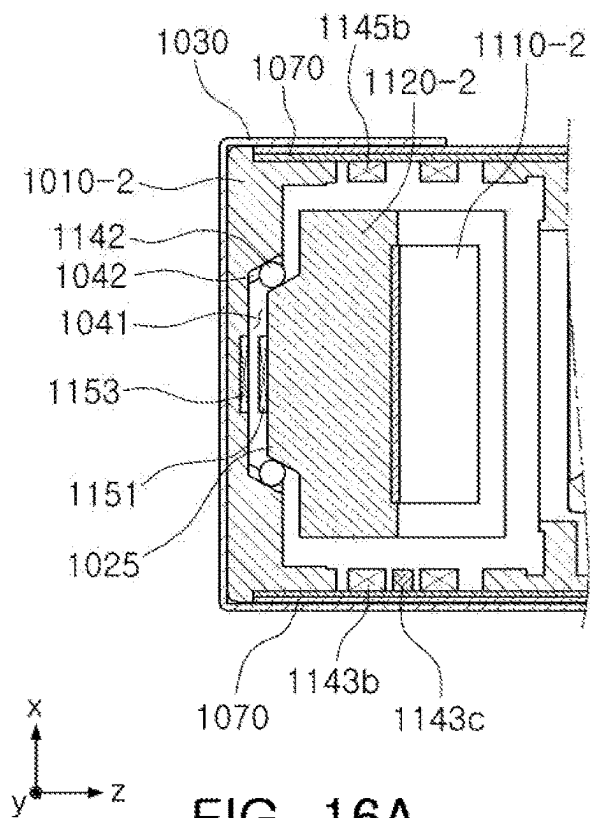


FIG. 15C



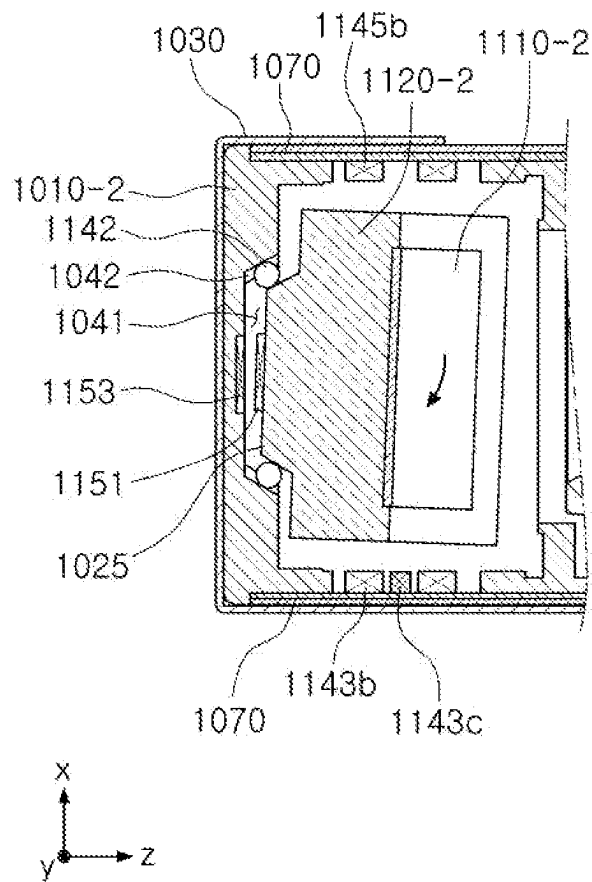


FIG. 16C

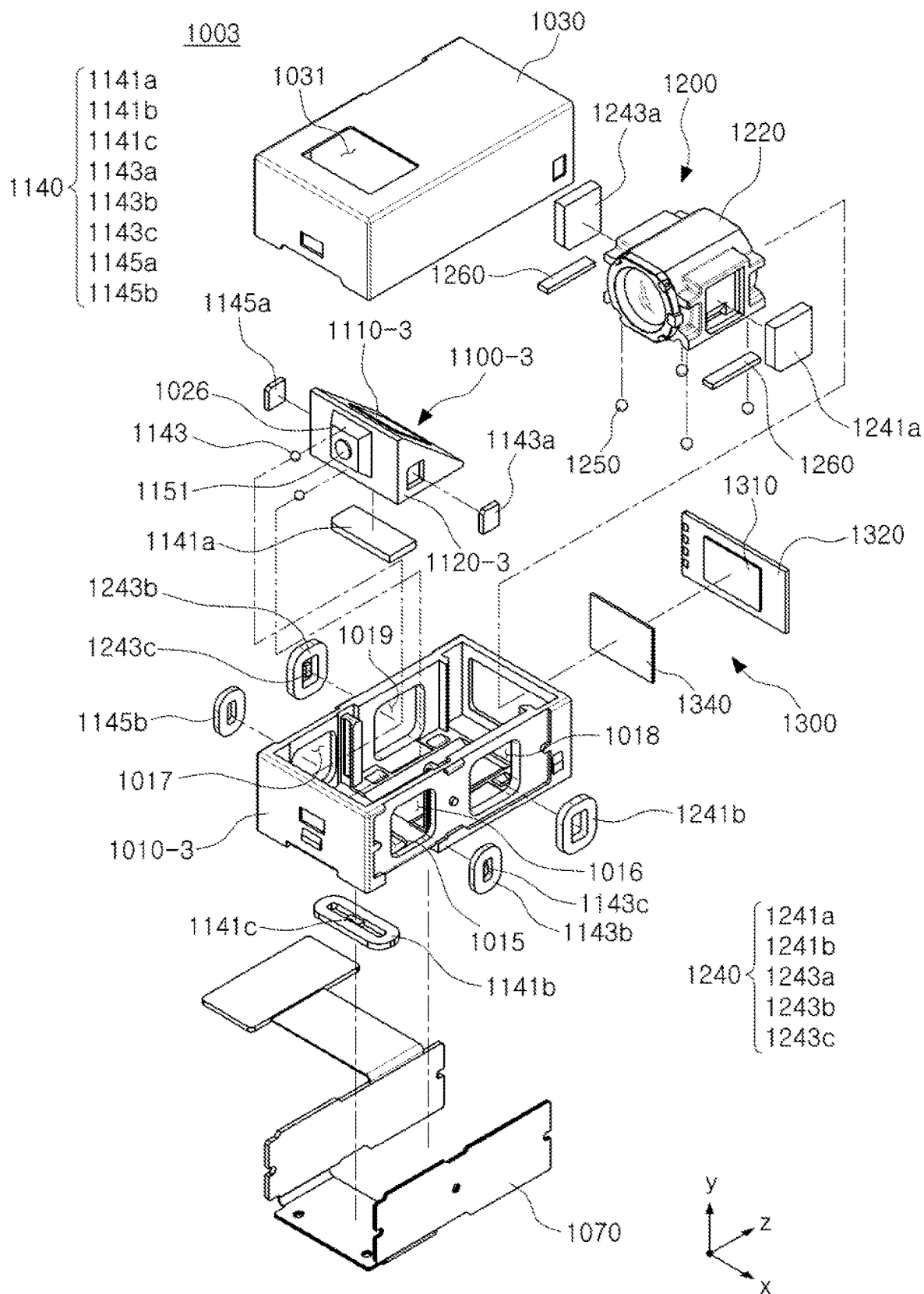


FIG. 17

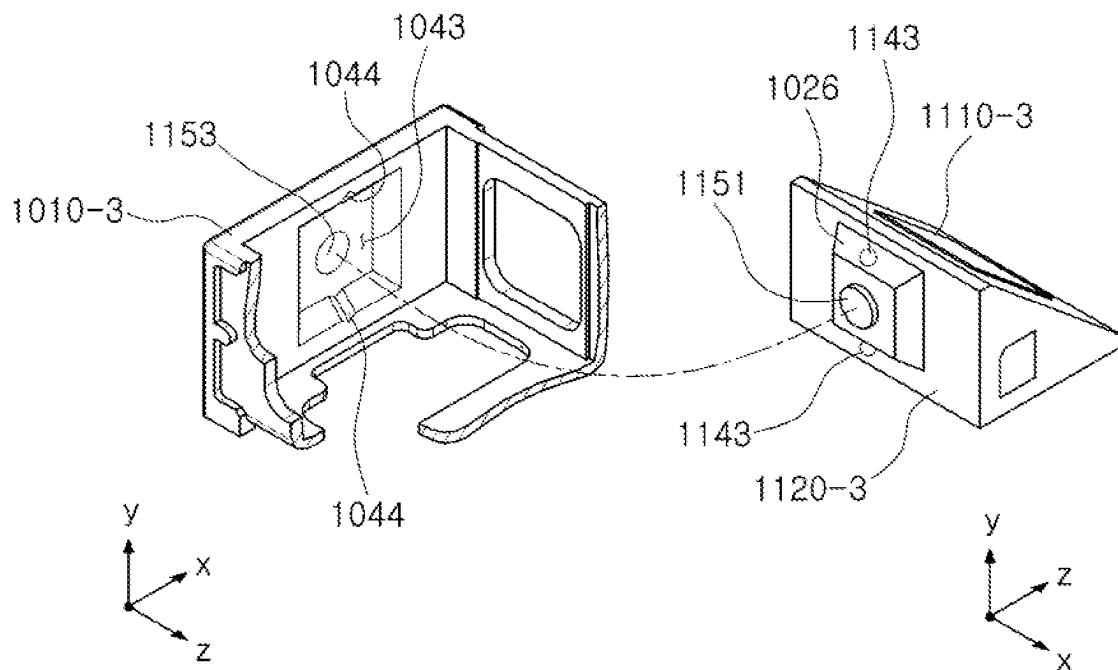


FIG. 18

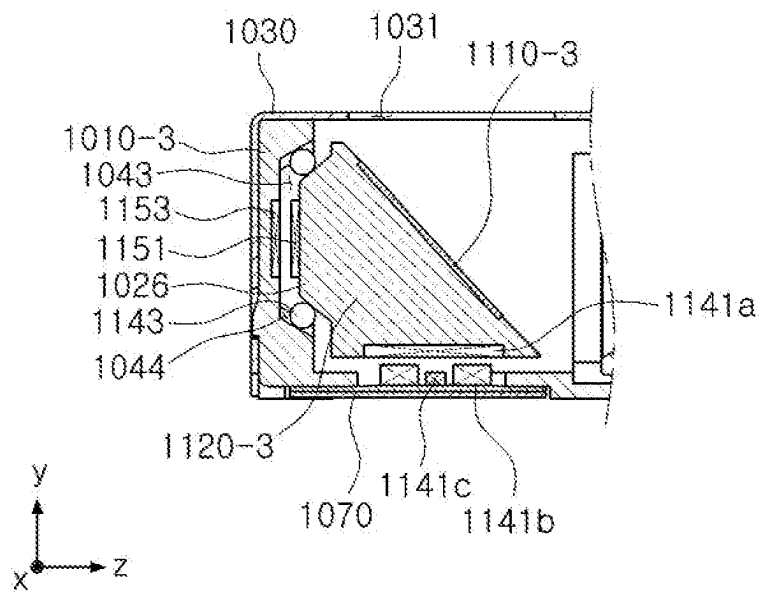


FIG. 19A

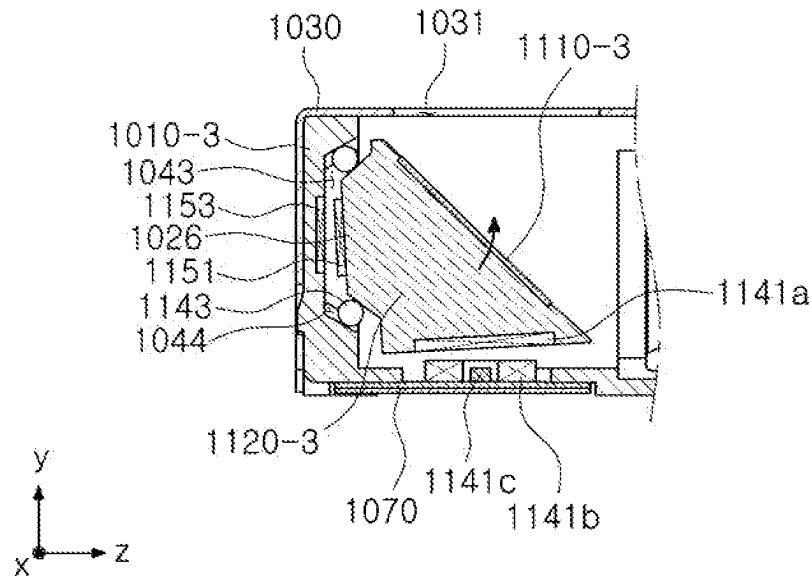


FIG. 19B

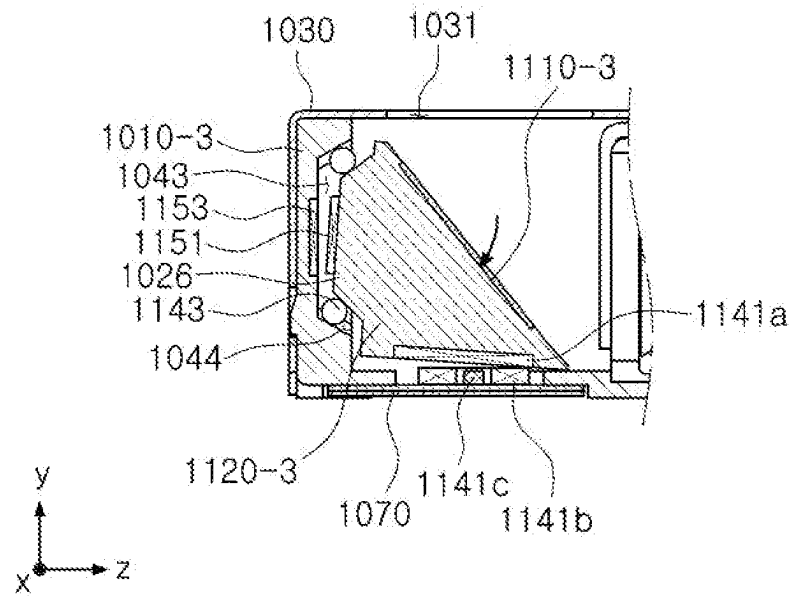


FIG. 19C

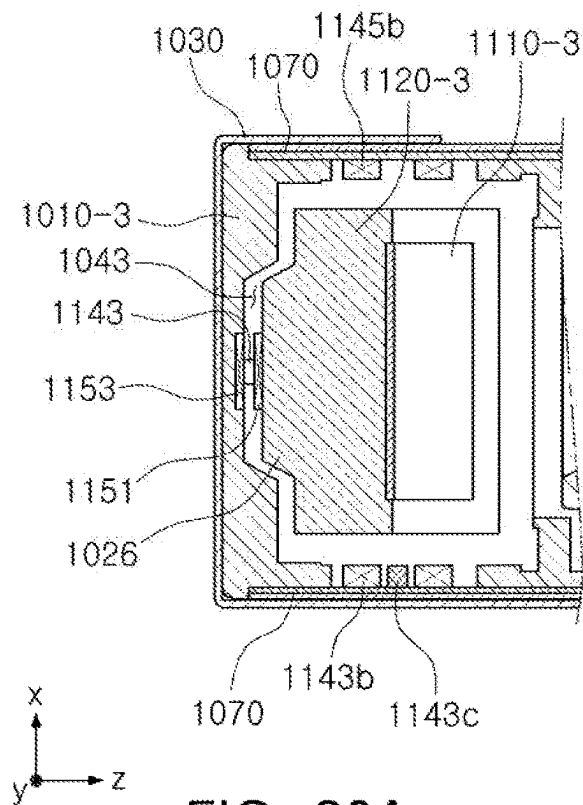


FIG. 20A

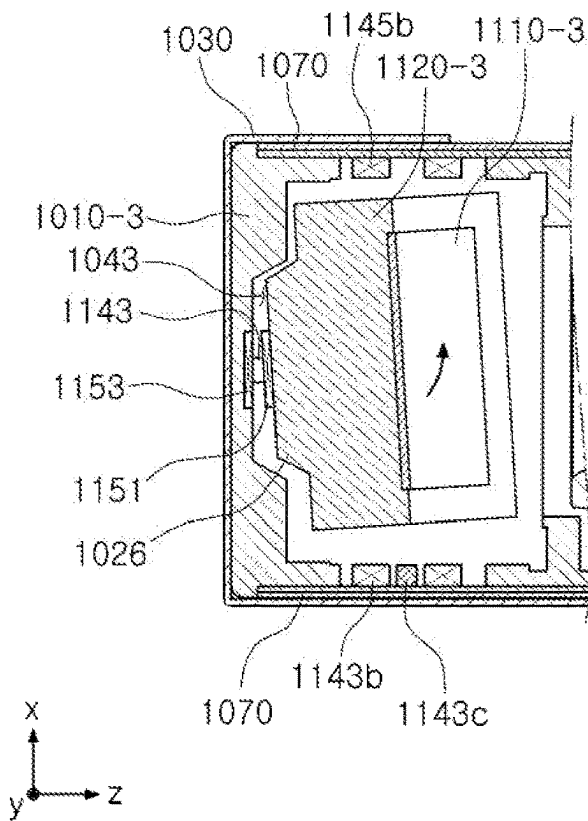


FIG. 20B

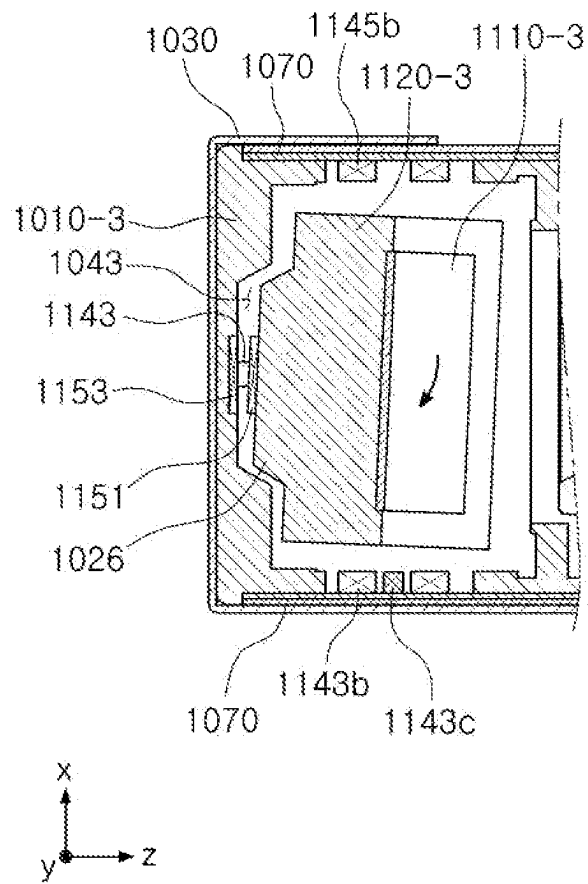


FIG. 20C

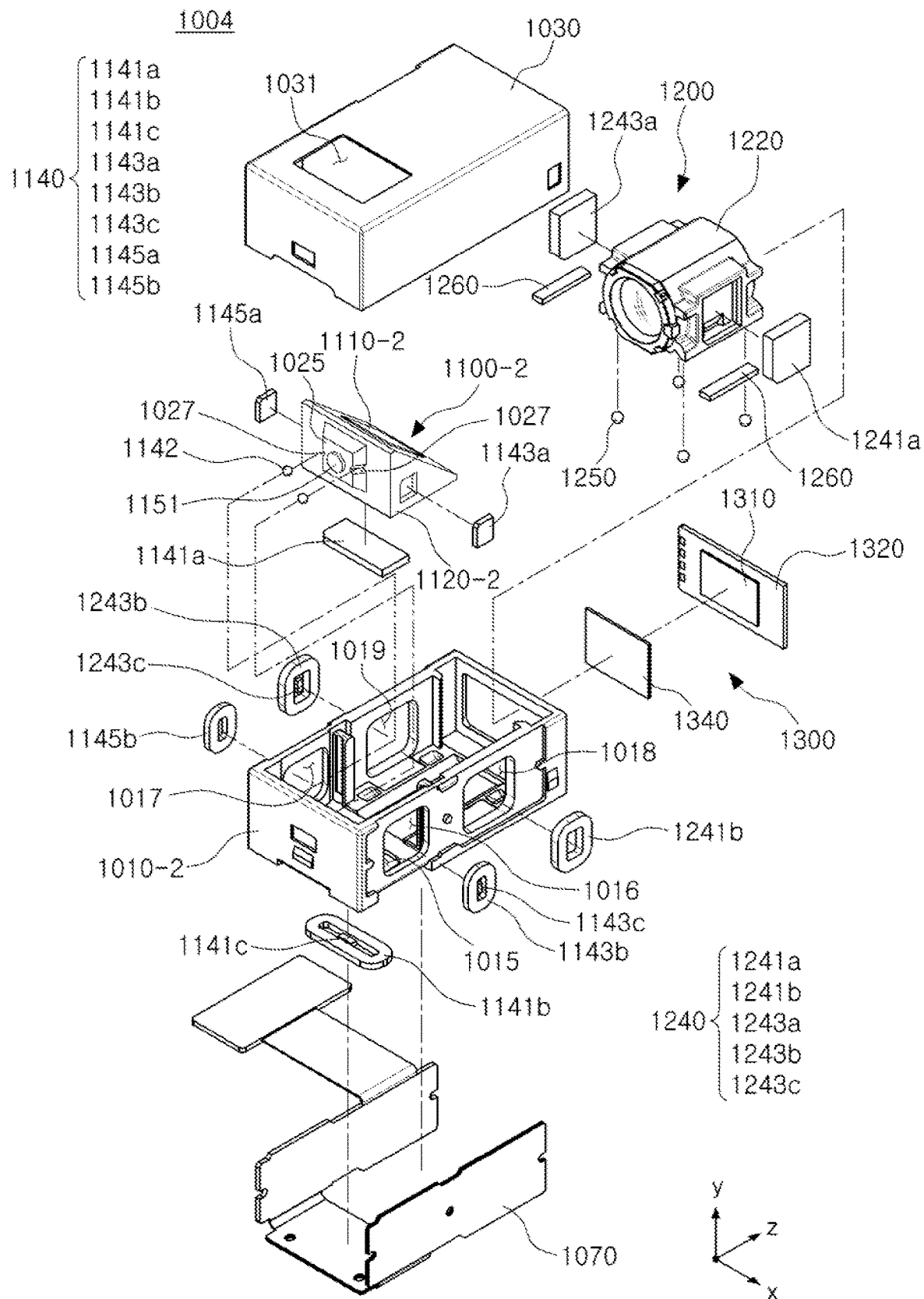


FIG. 21

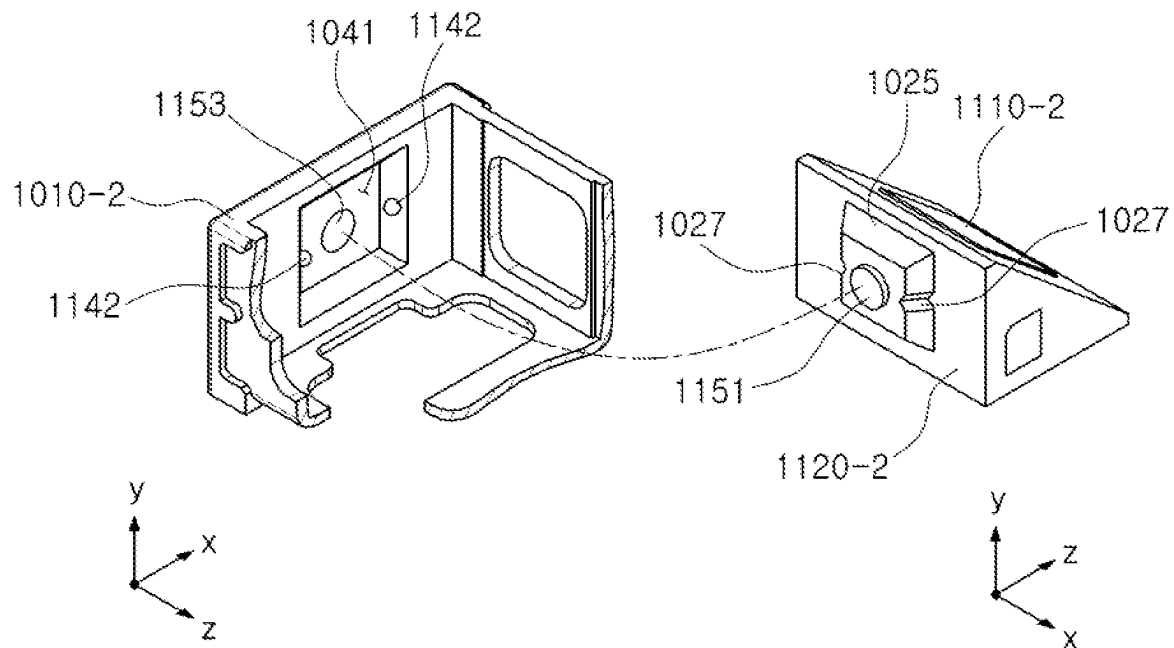


FIG. 22

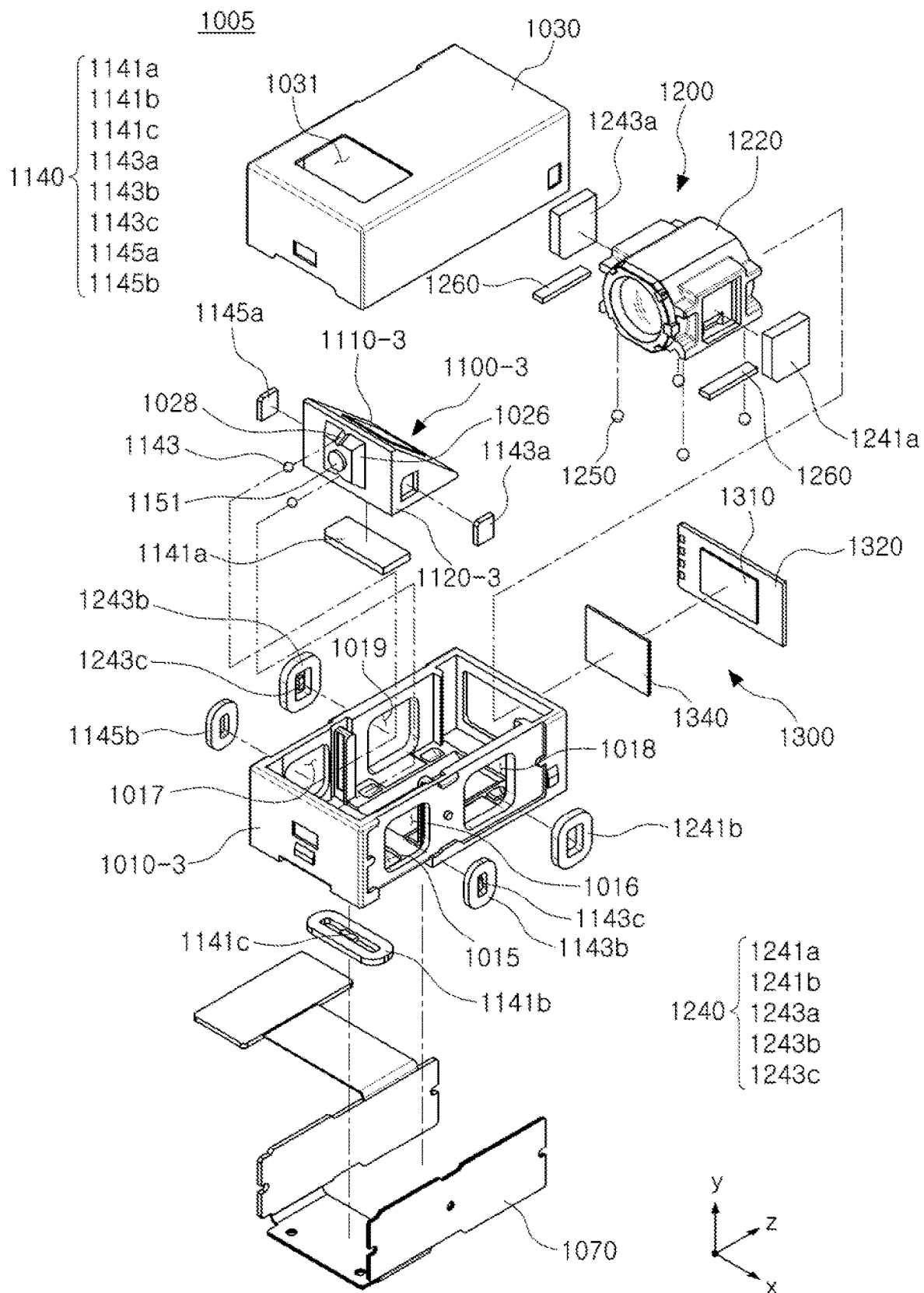


FIG. 23

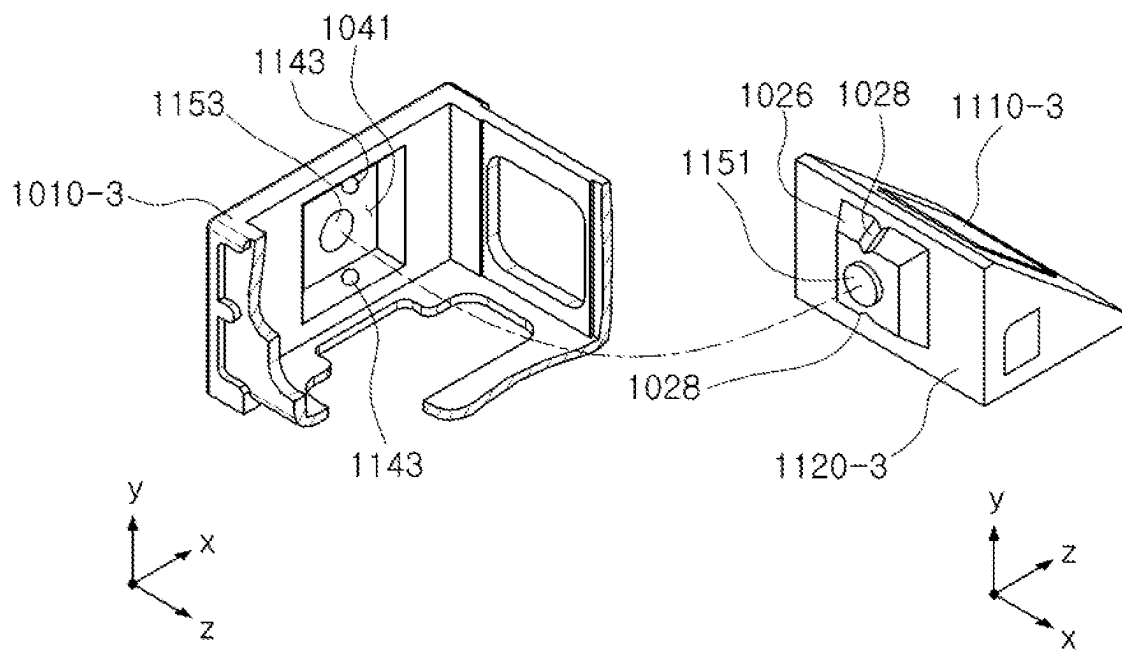


FIG. 24

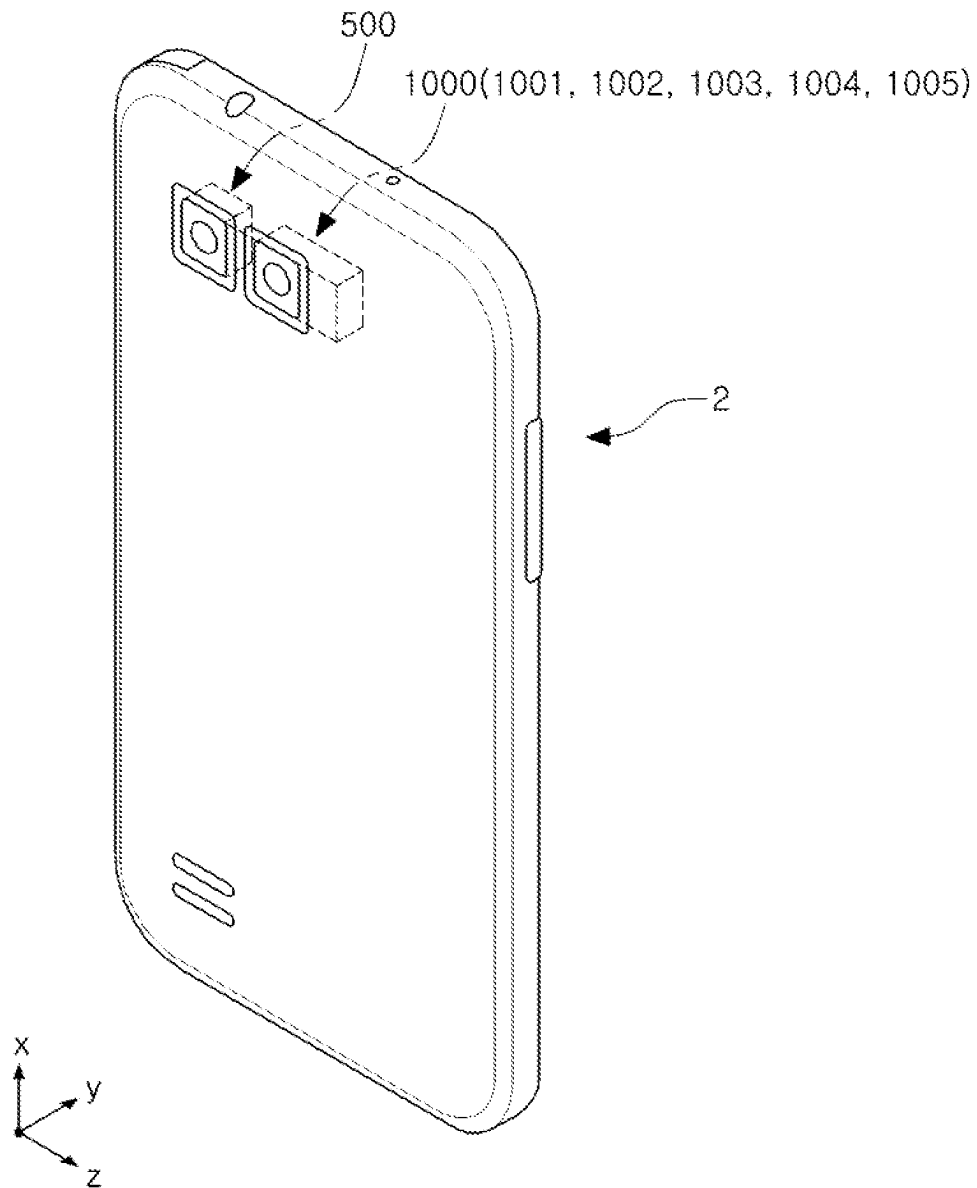


FIG. 25

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REFLECTING MODULE FOR OPTICAL IMAGE STABILIZATION (OIS) AND CAMERA MODULE INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/861,358 filed on Apr. 29, 2020, now U.S. Pat. No. 11,262,593 issued on Mar. 1, 2022, which is a continuation of U.S. patent application Ser. No. 15/854,578 filed on Dec. 26, 2017, now U.S. Pat. No. 10,678,062 issued on Jun. 9, 2020, which claims the benefit under 35 USC 119 (a) of Korean Patent Application Nos. 10-2017-0017527 filed on Feb. 8, 2017, and 10-2017-0049048 filed on Apr. 17, 2017, in the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference for all purposes.

BACKGROUND

1. Field

This application relates to a reflecting module for optical image stabilization (OIS) and a camera module including the same.

2. Description of Related Art

Camera modules have become a standard feature of portable electronic devices such as tablet personal computers (PCs), laptop PCs, and smartphones, and an autofocus function, an optical image stabilization (OIS) function, and a zoom function are some of the functions that have been implemented in camera modules for mobile terminals.

However, in order to implement such functions, the structures of such camera modules have become relatively complicated and the sizes thereof have increased, making it difficult to mount such camera modules in portable electronic devices.

In addition, when a lens or an image sensor is directly moved for the purpose of optical image stabilization, both a weight of the lens or of the image sensor itself and weights of other components to which the lens or the image sensor is attached should be considered, and a predetermined level or more of driving force is thus required, resulting in increased power consumption.

SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

In one general aspect, a reflecting module for optical image stabilization (OIS) includes a housing; a rotation holder provided in the housing and including a reflecting member; a rotation plate provided in the housing between an inner wall of the housing and the rotation holder so that the rotation holder is supported by the inner wall of the housing via the rotation plate; and a driving part configured to apply a driving force to the rotation holder to move the rotation holder.

The rotation plate may be rotatable with respect to the housing around a first axis substantially perpendicular to an

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optical axis, and the rotation holder may be rotatable with respect to the rotation plate around a second axis substantially perpendicular to the optical axis and the first axis.

In response to the rotation plate being rotated around the first axis, the rotation holder may rotate with the rotation plate, and in response to the rotation holder being rotated around the second axis, the rotation holder may rotate relative to the rotation plate.

The rotation plate may include first balls aligned in a first axis direction provided on a surface of the rotation plate facing the inner wall of the housing; and second balls aligned in a second axis direction perpendicular to the first axis direction provided on a surface of the rotation plate facing the rotation holder.

The reflecting module for OIS may further include first bearings aligned along a first axis substantially perpendicular to an optical axis provided between the rotation plate and the housing; and second bearings aligned along a second axis substantially perpendicular to the optical axis and the first axis provided between the rotation holder and the rotation plate.

The rotation plate may be rotatable around the first axis, and the rotation holder may be rotatable around the second axis.

The first bearings may be fixed to, or freely movable relative to, the rotation plate or the housing, and the second bearings may be fixed to, or freely movable relative to, the rotation plate or the rotation holder.

The first bearings and the second bearings may have a spherical shape or a spherical cap shape obtained by cutting a sphere with a plane.

The reflecting module for OIS may further include at least one first bearing having a cylindrical shape or a semi-cylindrical shape provided between the rotation plate and the housing and elongated along a first axis substantially perpendicular to an optical axis; and at least one second bearing having a cylindrical shape or a semi-cylindrical shape provided between the rotation holder and the rotation plate and elongated along a second axis substantially perpendicular to the optical axis and the first axis.

The reflecting module for OIS may further include a pulling magnet provided on one of the housing and the rotation holder; and a pulling yoke provided on a remaining one of the housing and the rotation holder; wherein the rotation holder may be supported by the inner wall of the housing via the rotation plate by an attractive force between the pulling magnet and the pulling yoke.

In another general aspect, a camera module includes a lens module including lenses; and the reflecting module for OIS described above provided in front of the lens module to change a path of light incident into the reflecting module for OIS so that the light having the changed path is directed toward the lens module.

In another general aspect, a reflecting module for optical image stabilization (OIS) includes a housing; a rotation holder provided in the housing and including a reflecting member and being supported by an inner wall of the housing; bearings aligned in a first axis direction substantially perpendicular to an optical axis direction provided between the housing and the rotation holder; and a driving part configured to apply a driving force to the rotation holder to move the rotation holder.

The bearings may be balls.

The rotation holder may be rotatable around substantially a first axis, and the rotation holder may be rotatable around a second axis perpendicular to an optical axis and the first axis.

In response to the rotation holder being rotated around the second axis, the rotation holder may slide on surfaces of the bearings.

In response to the rotation holder being rotated around the second axis, a rotation axis of the rotation holder may be formed at a portion spaced apart from the bearings toward the rotation holder by a predetermined distance.

In response to the rotation holder being rotated around the second axis, the rotation holder may be moved from a contact portion between the rotation holder and the bearings in the first axis direction.

The reflecting module for OIS may further include a pulling magnet provided on one of the housing and the rotation holder; and a pulling yoke provided on a remaining one of the housing and the rotation holder; wherein the rotation holder may be supported by the inner wall of the housing by an attractive force between the pulling magnet and the pulling yoke.

The housing or the rotation holder may be provided with seating grooves, and the bearings may be fixed to, or freely movable relative to, the housing or the rotation holder provided with the seating grooves.

A rotation protrusion may be provided on one of a surface of the housing facing the rotation holder and a surface of the rotation holder facing the housing, a rotation recess may be provided in a remaining one of the surface of the housing facing the rotation holder and the surface of the rotation holder facing the housing, and the bearings may be provided between the rotation protrusion and the rotation recess.

Edges of the rotation protrusion and the rotation recess may be inclined surfaces or round surfaces.

The rotation protrusion or the rotation recess may be provided with seating grooves elongated in a direction substantially perpendicular to a direction in which the bearings are arranged, and the bearings may be disposed in the seating grooves.

A cross section of the seating grooves may have a round shape or a polygonal shape.

In another general aspect, a camera module includes a lens module including lenses; and the reflecting module for OIS described above provided in front of the lens module to change a path of light incident into the reflecting module for OIS so that the light having the changed path is directed toward the lens module.

In another general aspect, a reflecting module for optical image stabilization (OIS) module includes a housing; a rotation holder supported in the housing by friction; a reflecting member disposed on the rotation holder; and a driving part configured to rotate the rotation holder relative to the housing.

The reflecting module for OIS may further include a rotation plate disposed between an inner wall of the housing and the rotation holder; first bearings arranged in a direction of a first axis and disposed between the inner wall of the housing and the rotation plate; and second bearings arranged in a direction of a second axis substantially perpendicular to the first axis and disposed between the rotation plate and the rotation holder; wherein the driving part may be further configured to rotate the rotation plate around the first axis, and rotate the rotation holder around the second axis.

The reflecting module for OIS may further include a rotation recess provided in one of an inner wall of the housing facing the rotation holder and a surface of the rotation holder facing the inner wall of the housing; and a rotation protrusion provided on a remaining one of the inner wall of the housing and the surface of the rotation holder facing the inner wall of the housing and engaging the

rotation recess; wherein the rotation recess and the rotation protrusion have respective shapes that limit rotation of the rotation holder to rotation around a first axis and a second axis substantially perpendicular to the first axis, and the driving part may be further configured to rotate the rotation holder about the first axis and the second axis.

The reflecting module for OIS may further include seating grooves provided in one of a surface of the rotation recess facing the rotation protrusion and a surface of the rotation protrusion facing the rotation recess; and bearings arranged in a direction of one of the first axis and the second axis and disposed in the seating grooves; wherein the seating grooves are elongated in a direction of a third axis substantially perpendicular to the first axis and the second axis.

In another general aspect, a reflecting module for optical image stabilization (OIS) includes a housing; a rotation holder disposed in the housing; a reflecting member disposed on the rotation holder; a driving part configured to rotate the rotation holder relative to the housing; a pulling magnet disposed on one of the housing and the rotation holder; and a pulling yoke disposed on a remaining one of the housing and the rotation holder; wherein the rotation holder is pressed against an inner wall of the housing by an attractive force between the pulling magnet and the pulling yoke.

The reflecting module for OIS may further include a rotation plate disposed between the inner wall of the housing and the rotation holder; first bearings arranged in a direction of a first axis and disposed between the inner wall of the housing and the rotation plate; and second bearings arranged in a direction of a second axis substantially perpendicular to the first axis and disposed between the rotation plate and the rotation holder; wherein the driving part may be further configured to rotate the rotation plate around the first axis, and rotate the rotation holder around the second axis.

The reflecting module for OIS may further include a rotation recess provided in one of the inner wall of the housing and a surface of the rotation holder facing the inner wall of the housing; and a rotation protrusion provided on a remaining one of the inner wall of the housing and the surface of the rotation holder facing the inner wall of the housing and engaging the rotation recess; wherein the rotation recess and the rotation protrusion have respective shapes that limit rotation of the rotation holder to rotation around a first axis and a second axis substantially perpendicular to the first axis, and the driving part may be further configured to rotate the rotation holder about the first axis and the second axis.

The reflecting module for OIS may further include seating grooves provided in one of a surface of the rotation recess facing the rotation protrusion and a surface of the rotation protrusion facing the rotation recess; and bearings arranged in a direction of one of the first axis and the second axis and disposed in the seating grooves; wherein the seating grooves are elongated in a direction of a third axis substantially perpendicular to the first axis and the second axis.

Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view illustrating an example of a portable electronic device.

FIG. 2 is a perspective view illustrating an example of a camera module.

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FIGS. 3A and 3B are cross-sectional views taken along the lines IIIA-III A' and IIIB-IIIB' in FIG. 2 illustrating the camera module of FIG. 2.

FIG. 4 is an exploded perspective view illustrating the camera module of FIG. 2.

FIG. 5 is a perspective view illustrating a housing of the camera module of FIG. 2.

FIGS. 6A and 6B are exploded perspective views illustrating examples of a rotation plate and a rotation holder of the camera module of FIG. 2.

FIG. 7 is a perspective view illustrating a lens holder of the camera module of FIG. 2.

FIG. 8 is an assembled perspective view illustrating components other than a cover in the camera module of FIG. 2.

FIG. 9 is an assembled perspective view illustrating the housing and a board in the camera module of FIG. 2.

FIG. 10 is an exploded perspective view illustrating the housing and the rotation holder in the camera module of FIG. 2.

FIGS. 11A through 11C are cross-sectional views taken along the lines XIA-XIA' through XIC-XIC' in FIG. 10 illustrating how the rotation holder of the camera module of FIG. 2 is rotated around a first axis.

FIGS. 12A through 12C are cross-sectional views taken along the lines XIIA-XIIA' through XIIC-XIIC' in FIG. 10 illustrating how the rotation holder of the camera module of FIG. 2 is rotated around a second axis.

FIG. 13 is an exploded perspective view illustrating another example of a camera module.

FIG. 14 is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. 13.

FIGS. 15A through 15C are cross-sectional views illustrating how the rotation holder of the camera module of FIG. 13 is rotated around a first axis.

FIGS. 16A through 16C are cross-sectional views illustrating how the rotation holder of the camera module of FIG. 13 is rotated around a second axis.

FIG. 17 is an exploded perspective view illustrating another example of a camera module.

FIG. 18 is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. 17.

FIGS. 19A through 19C are cross-sectional views illustrating how the rotation holder of the camera module of FIG. 18 is rotated around a first axis.

FIGS. 20A through 20C are cross-sectional views illustrating how the rotation holder of the camera module of FIG. 18 is rotated around a second axis.

FIG. 21 is an exploded perspective view illustrating another example of a camera module.

FIG. 22 is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. 21.

FIG. 23 is an exploded perspective view illustrating another example of a camera module.

FIG. 24 is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. 23.

FIG. 25 is a perspective view illustrating another example of a portable electronic device.

Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

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DETAILED DESCRIPTION

The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that are known in the art may be omitted for increased clarity and conciseness.

The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.

Throughout the specification, when an element, such as a layer, region, or substrate, is described as being "on," "connected to," or "coupled to" another element, it may be directly "on," "connected to," or "coupled to" the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being "directly on," "directly connected to," or "directly coupled to" another element, there can be no other elements intervening therebetween.

Although terms such as "first," "second," and "third" may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

Spatially relative terms such as "above," "upper," "below," and "lower" may be used herein for ease of description to describe one element's relationship to another element as shown in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being "above" or "upper" relative to another element will then be "below" or "lower" relative to the other element. Thus, the term "above" encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

FIG. 1 is a perspective view illustrating an example of a portable electronic device.

Referring to FIG. 1, a portable electronic device 1 may be a portable electronic device such as a mobile communications terminal, a smartphone, or a tablet personal computer (PC) in which a camera module 1000, 1001, 1002, 1003, 1004, or 1005 is mounted.

As illustrated in FIG. 1, the portable electronic device 1 includes the camera module 1000 to capture an image of a subject.

In this example, the camera module 1000, 1001, 1002, 1003, 1004, or 1005 includes lenses, and an optical axis (a Z axis) of each of the lenses is oriented in a direction perpendicular to a thickness direction of the portable electronic device 1 (a Y-axis direction or a direction from a front surface of the portable electronic device 1 to a rear surface thereof, or an opposite direction from the rear surface of the portable electronic device 1 to the front surface thereof).

For example, the optical axis (the Z axis) of each of the lenses included in the camera module 1000 may be formed in a width direction or a length direction (an X-axis direction or a Z-axis direction) of the portable electronic device 1.

Therefore, even in the case that the camera module 1000 has functions such as an autofocus (AF) function, a zoom function, and an optical image stabilization (OIS) function, a thickness of the portable electronic device 1 will not be increased. Therefore, the portable electronic device 1 may be miniaturized.

In this example, the camera module 1000 has any one or any combination of any two or more of the AF function, the zoom function, and the OIS function.

Since the camera module 1000 including the AF function, the zoom function, and the OIS function needs additional components to implement these functions, a size of the camera module will be increased compared to a camera module that does not include these functions.

When the size of the camera module 1000 is increased, it may be difficult to miniaturize the portable electronic device 1 in which the camera module 1000 is mounted.

For example, when the number of stacked lenses in the camera module is increased for the purpose of the zoom function and the stacked lenses are provided in the camera module in the thickness direction of the portable electronic device 1, a thickness of the portable electronic device may also be increased depending on the number of stacked lenses. Therefore, when the thickness of the portable electronic device 1 is not increased, the number of stacked lenses is limited by the thickness of the portable electronic device 1, such that a zoom performance may be deteriorated.

In addition, an actuator moving a lens group in an optical axis direction or a direction perpendicular to the optical axis needs to be installed in order to implement the AF function and the OIS function, and when the optical axis (the Z axis) of the lens group is oriented in the thickness direction of the portable electronic device 1, the actuator moving the lens group also needs to be installed in the thickness direction of the portable electronic device 1. Therefore, a thickness of the portable electronic device 1 is increased.

However, in the camera module 1000 of this example, the optical axis (the Z axis) of each of the lenses is oriented perpendicularly to the thickness direction of the portable electronic device 1. Therefore, even in the case that the camera module 1000 having the AF function, the zoom function, and the OIS function is mounted in the portable electronic device 1, the portable electronic device 1 may be miniaturized.

FIG. 2 is a perspective view illustrating an example of a camera module, and FIGS. 3A and 3B are cross-sectional views taken along the lines IIIA-III A' and IIIB-IIIB' in FIG. 2 illustrating the camera module of FIG. 2.

Referring to FIGS. 2 through 3B, a camera module 1001 includes a reflecting module 1100, a lens module 1200, and an image sensor module 1300 provided in a housing 1010.

The reflecting module 1100 changes a path of light. For example, a path of light incident through an opening 1031 (see FIG. 3A) of a cover 1030 covering an upper portion of the camera module 1001 is changed by the reflecting module 1100 so that the light is directed toward the lens module 1200. To this end, the reflecting module 1100 includes a reflecting member 1110 reflecting the light.

A path of the light incident through the opening 1031 is changed by the reflecting module 1100 so that the light is directed toward the lens module 1200. For example, a path of light incident in the thickness direction (the Y-axis direction) of the camera module 1001 is changed by the reflecting module 1100 to substantially coincide with the optical axis direction (the Z-axis direction).

The lens module 1200 includes lenses through which the light having the path changed by the reflecting module 1100 passes. The image sensor module 1300 includes an image sensor 1310 converting the light passing through the lenses into an electrical signal, a printed circuit board 1320 on which the image sensor 1310 is mounted, and an optical filter 1340 filtering the light incident thereon from the lens module 1200. The optical filter 1340 may be an infrared cut-off filter.

In an internal space of the housing 1010, the reflecting module 1100 is provided in front of the lens module 1200, and the image sensor module 1300 is provided behind the lens module 1200.

Referring to FIGS. 2 through 10, the camera module 1001 includes the reflecting module 1100, the lens module 1200, and the image sensor module 1300 provided in the housing 1010.

The reflecting module 1100, the lens module 1200, and the image sensor module 1300 are sequentially provided in the housing 1010 from one side of the housing 1010 to the other side of the housing 1010. The housing 1010 has the internal space in which the reflecting module 1100, the lens module 1200, and the image sensor module 1300 are provided. In this example, the printed circuit board 1320 included in the image sensor module 1300 is attached to an outer surface of the housing 1010. In this example, the housing 1010 is a single housing and both of the reflecting module 1100 and the lens module 1200 are provided in the internal space of the housing 1010. However, the housing 1010 is not limited thereto. For example, two separate housings in which the reflecting module 1100 and the lens module 1200 are respectively provided may be connected to each other.

In addition, the housing 1010 is covered by the cover 1030 so that the internal space of the housing 1010 is not visible.

The cover 1030 has the opening 1031 through which light is incident, and a path of the light incident through the opening 1031 is changed by the reflecting module 1110 so that the light is incident on the lens module 1200. In the example illustrated in the drawings, the cover 1030 is a single cover covering the entire housing 1010. In another example, two separate covers respectively covering the reflecting module 1100 and the lens module 1200 may be provided.

The reflecting module 1100 includes the reflecting member 1110 reflecting the light. In addition, the light incident to the lens module 1200 passes through the lenses and is then converted into the electrical signal by the image sensor 1310.

The housing 1010 includes the reflecting module 1100 and the lens module 1200 provided in the internal space thereof. Therefore, in the internal space of the housing 1010,

a space in which the reflecting module **1100** is provided and a space in which the lens module **1200** is provided are delineated from each other by protruding walls **1007**. In addition, the reflecting module **1100** is provided in front of the protruding walls **1007**, and the lens module **1200** is provided behind the protruding walls **1007**. The protruding walls **1007** protrude from opposite sidewalls of the housing **1010** into the internal space.

The reflecting module **1100** provided in front of the protruding walls **1007** has a structure in which a rotation holder **1120** is held against and supported by an inner wall surface of the housing **1010** by an attractive force between a pulling yoke **1153** provided on the inner wall surface of the housing **1010** and a pulling magnet **1151** provided on the rotation holder **1120**. Alternatively, although not illustrated in the drawings, the pulling magnet **1151** may be provided on the inner wall surface of the housing **1010**, and the pulling yoke **1153** may be provided on the rotation holder **1120**. However, the structure illustrated in the drawings will hereinafter be described for convenience of explanation.

In addition, first ball bearings **1131**, a rotation plate **1130**, and second ball bearings **1133** are provided between the inner wall surface of the housing **1010** and the rotation holder **1120**. Since the first ball bearings **1131** and the second ball bearings **1133** are held in the seating grooves **1132**, **1134**, **1021**, and **1121** while being partially inserted into the seating grooves **1132**, **1134**, **1021**, and **1121** as described below, when the rotation holder **1120** and the rotation plate **1130** are inserted into the internal space of the housing **1010**, a slight space may need to be provided between the rotation holder **1120** and the protruding walls **1007**, and after the rotation holder **1120** is mounted in the housing **1010**, the rotation holder **1120** is held against the inner wall surface of the housing **1010** and the first ball bearings **1131** and the second ball bearings **1133** are held in the seating grooves **1132**, **1134**, **1021**, and **1121** by the attractive force between the pulling yoke **1153** and the pulling magnet **1151**, and a slight space may thus remain between the rotation holder **1120** and the protruding walls **1007**.

Therefore, in this example, the housing **1010** includes stoppers **1050** fitted onto the protruding walls **1007** while supporting the rotation holder **1120** and having a hook shape (even in the case that the stoppers **1050** are not provided, the rotation holder **1120** may be fixed to the housing by the attractive force between the pulling magnet **1151** and the pulling yoke **1153**). In this example, the stoppers **1050** have the hook shape, and support the rotation holder **1120** in a state in which hook portions thereof are hooked onto the protruding walls **1007**. The stoppers **1050** serve as brackets supporting the rotation holder **1120** when the reflecting module **1100** is not driven, and additionally serve as the stoppers **1050** limiting movement of the rotation holder **1120** when the reflecting module **1100** is driven. The stoppers **1050** are respectively provided on the protruding walls **1007** protruding from the opposite sidewalls of the housing. A space is provided between the stoppers **1050** and the rotation holder **1120** so that the rotation holder **1120** is smoothly rotated. Alternatively, the stoppers **1050** may be formed of an elastic material to allow the rotation holder **1120** to be smoothly moved in a state in which the rotation holder **1120** is supported by the stoppers **1050**.

In addition, the housing **1010** includes a first driving part **1140** and a second driving part **1240** provided to respectively drive the reflecting module **1100** and the lens module **1200**. The first driving part **1140** includes coils **1141b**, **1143b**, and **1145b** for driving the reflecting module **1100**,

and the second driving part **1240** includes coils **1241b** and **1243b** for driving the lens module **1200**. In addition, since the coils **1141b**, **1143b**, **1145b**, **1241b**, and **1243b** are provided in the housing **1010** in a state in which they are mounted on a main board **1070**, the housing **1010** is provided with through-holes **1015**, **1016**, **1017**, **1018**, and **1019** so that the coils **1141b**, **1143b**, **1145b**, **1241b**, and **1243b** are exposed to the internal space of the housing **1010**.

In the example illustrated in the drawings, the main board **1070** on which the coils **1141b**, **1143b**, **1145b**, **1241b**, and **1243b** are mounted is a single board. In this case, one set of terminals may be provided, and connection of an external power supply and signals may thus be easy. However, the main board **1070** is not limited thereto, but may also be provided as two boards by separating a board on which the coils **1141b**, **1143b**, and **1145b** of the reflecting module **1100** are mounted from a board on which the coils **1241b** and **1243b** of the lens module **1200** are mounted.

The reflecting module **1100** changes the path of the light incident thereto through the opening **1031**. When a still image or a moving image is captured, the still image may be blurred or the moving image may be shaken due to a hand-shake or other movement of a user. In this case, the reflecting module **1100** corrects the hand-shake or other movement of the user by moving the rotation holder **1120** on which the reflecting member **1110** is mounted to perform an OIS function. For example, when shaking is generated at the time of capturing the still image or the moving image due to the hand-shake or other movement of the user, a relative displacement corresponding to the shaking may be provided to the rotation holder **1120** to compensate for the shaking.

In this example, the OIS function is implemented by the movement of the rotation holder **1120** having a relatively low weight since it does not include lenses or other components, and thus power consumption for the OIS function is significantly reduced.

That is, in this example, the light on which the OIS is performed is incident to the lens module **1200** by changing the path of the light by the movement of the rotation holder **1120** on which the reflecting member **1110** is provided without moving a lens holder including the lenses or the image sensor in order to implement the OIS function.

The reflecting module **1100** includes the rotation holder **1120** provided in the housing **1010** and supported by the housing **1010**, the reflecting member **1110** mounted on the rotation holder **1120**, and the first driving part **1140** moving the rotation holder **1120**.

The reflecting member **1110** changes the path of the light. For example, the reflecting member **1110** may be a mirror or a prism reflecting the light (a case in which the reflecting member **1110** is the prism is illustrated in the drawings associated with the camera module **1001** of FIGS. 2 through 10 for convenience of explanation).

The reflecting member **1110** is fixed to the rotation holder **1120**. The rotation holder **1120** has a mounting surface **1123** on which the reflecting member **1110** is mounted.

The mounting surface **1123** of the rotation holder **1120** is an inclined surface so that the path of the light is changed. In one example, the mounting surface **1123** is an inclined surface inclined with respect to the optical axis (the Z axis) of each of the lenses by 30° to 60°. In addition, the inclined surface of the rotation holder **1120** is directed toward the opening **1031** of the cover **1030** through which the light is incident.

The rotation holder **1120** on which the reflecting member **1110** is mounted is movably mounted in the internal space of the housing **1010**. For example, the rotation holder **1120** is

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mounted in the housing 1010 to be rotatable around a first axis (an X axis) and a second axis (a Y axis). In this example, the first axis (the X axis) and the second axis (the Y axis) are axes that are perpendicular to the optical axis (the Z axis), and are perpendicular to each other.

The rotation holder 1120 is supported in the housing 1010 by the first ball bearings 1131 aligned along the first axis (the X axis) and the second ball bearings 1133 aligned along the second axis (the Y axis) so that the rotation holder 1120 can be smoothly rotated around the first axis (the X axis) and the second axis (the Y axis). In the drawings, two first ball bearings 1131 aligned along the first axis (the X axis) and two second ball bearings 1133 aligned along the second axis (the Y axis) are illustrated by way of example. In addition, the rotation holder 1120 is rotated around the first axis (the X axis) and the second axis (the Y axis) by a first driving part 1140 to be described below.

In this example, unlike other examples described below with respect to FIGS. 13-24, since the first ball bearings 1131 aligned along the first axis (the X axis) and the second ball bearings 1133 aligned along the second axis (the Y axis) are provided, the two first ball bearings 1131 aligned along the first axis (the X axis) may be provided in a cylindrical shape extending in the first axis direction (the X axis direction), and the two second ball bearings 1133 aligned along the second axis (the Y axis) may be provided in a cylindrical shape extending in the second axis direction (the Y axis direction). In this case, the seating grooves 1132, 1134, 1021, and 1121 may be provided in a semi-cylindrical shape corresponding to shapes of the first and second ball bearings (see FIG. 6B). Meanwhile, a case in which the two first ball bearings 1131 and the two second ball bearings 1133 are provided is illustrated in FIG. 6B, but one or two or more first ball bearings 1131 and second ball bearings 1133 elongated in the first axis direction (the X-axis direction) or the second axis direction (the Y-axis direction) may also be provided.

In the example illustrated in FIGS. 2-10, the first ball bearings 1131 and the second ball bearings 1133 are provided on a front surface and a rear surface of the rotation plate 1130, respectively. In another example, the first ball bearings 1131 and the second ball bearings 1133 are provided on a rear surface and a front surface of the rotation plate 1130, respectively. That is, the first ball bearings 1131 may be aligned along the second axis (the Y axis) and the second ball bearings 1133 may be aligned along the first axis (the X axis). However, the structure illustrated in the drawings will hereinafter be described for convenience of explanation. The rotation plate 1130 is provided between the rotation holder 1120 and an inner surface of the housing 1010. In addition, the rotation holder 1120 is supported by the housing 1010 through the rotation plate 1130, the first ball bearings 1131, and the second ball bearings 1133 by the attractive force between the pulling magnet 1151 or the pulling yoke provided on the rotation holder 1120 and the pulling yoke 1153 or the pulling magnet provided on the housing 1010.

The seating grooves 1132 and 1134 into which the first ball bearings 1131 and the second ball bearings 1133 are inserted, respectively, are provided in the front surface and the rear surface of the rotation plate 1130, respectively, and include first seating grooves 1132 into which the first ball bearings 1131 are partially inserted and second seating grooves 1134 into which the second ball bearings 1133 are partially inserted.

In addition, the housing 1010 is provided with third seating grooves 1021 into which the first ball bearings 1131

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are partially inserted, and the rotation holder 1120 is provided with fourth seating grooves 1121 into which the second ball bearings 1133 are partially inserted.

The first seating grooves 1132, the second seating grooves 1134, the third seating grooves 1021, and the fourth seating grooves 1121 described above may be provided in a hemispherical or polygonal (poly-prismatic or poly-pyramidal) groove shape so that the first ball bearings 1131 and the second ball bearings 1133 can easily rotate (depths of the seating grooves 1132, 1134, 1021, and 1121 may be smaller than radii thereof to enable the first ball bearings 1131 and the second ball bearings 1133 to be easily rotated. The first ball bearings 1131 and the second ball bearings 1133 are not entirely disposed inside the seating grooves, but are partially exposed to enable the rotation plate 1130 and the rotation holder 1120 to be easily rotated. In addition, positions and the numbers of first seating grooves 1132, second seating grooves 1134, third seating grooves 1021, and fourth seating grooves 1121 correspond to positions and numbers of the first ball bearings 1131 aligned along the first axis (the X axis) and the second ball bearings 1133 aligned along the second axis (the Y axis).

The first ball bearings 1131 and the second ball bearings 1133 serve as bearings while rolling in the first seating grooves 1132, the second seating grooves 1134, the third seating grooves 1021, and the fourth seating grooves 1121.

In another example, the first ball bearings 1131 and the second ball bearings 1133 have a structure in which they are fixedly provided in at least one of the housing 1010, the rotation plate 1130, and the rotation holder 1120. For example, the first ball bearings 1131 are fixedly provided in the housing 1010 or the rotation plate 1130, and the second ball bearings 1133 are fixedly provided in the rotation plate 1130 or the rotation holder 1120. In this case, only a member facing a member in which the first ball bearings 1131 or the second ball bearings 1133 are fixedly provided is provided with the seating grooves. In this case, the ball bearings serve as friction bearings by sliding in the seating grooves rather than rotating in the seating grooves.

When the first ball bearings 1131 and the second ball bearings 1133 are fixedly provided in any one of the housing 1010, the rotation plate 1130, and the rotation holder 1120, the first ball bearings 1131 and the second ball bearings 1133 may be provided in a spherical shape or a hemispherical shape. However, a case in which the first ball bearings 1131 and the second ball bearings 1133 are provided in a hemispherical shape is only an example, and the first ball bearings 1131 and the second ball bearings 1133 may also be provided to have a protruding length greater or smaller than that of a hemisphere. As described above, a case in which the ball bearings 1131 and 1133 are provided in cylindrical shapes extending along the first axis (the X axis) and the second axis (the Y axis), respectively, may also be similarly applied.

In addition, the first ball bearings 1131 and the second ball bearings 1133 may be separately manufactured and be then attached to any one of the housing 1010, the rotation plate 1130, and the rotation holder 1120. Alternatively, the first ball bearings 1131 and the second ball bearings 1133 may be provided integrally with the housing 1010, the rotation plate 1130, or the rotation holder 1120 at the time of manufacturing the housing 1010, the rotation plate 1130, or the rotation holder 1120.

The first driving part 1140 generates a driving force so that the rotation holder 1120 is rotatable around the two axes.

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As an example, the first driving part **1140** includes magnets **1141a**, **1143a**, and **1145a** and the coils **1141b**, **1143b**, and **1145b** disposed to face the magnets **1141a**, **1143a**, and **1145a**, respectively.

When power is applied to the coils **1141b**, **1143b**, and **1145b**, the rotation holder **1120** on which the magnets **1141a**, **1143a**, and **1145a** are mounted is rotated around the first axis (the X axis) and the second axis (the Y axis) by electromagnetic interaction between the magnets **1141a**, **1143a**, and **1145a** and the coils **1141b**, **1143b**, and **1145b**.

The magnets **1141a**, **1143a**, and **1145a** are mounted on the rotation holder **1120**. In the example illustrated in FIGS. 2-10, the magnet **1141a** of the magnets **1141a**, **1143a**, and **1145a** is mounted on a lower surface of the rotation holder **1120**, and the magnets **1143a** and **1145a** of the magnets **1141a**, **1143a**, and **1145a** are mounted on side surfaces of the rotation holder **1120**.

The coils **1141b**, **1143b**, and **1145b** are mounted on the housing **1010**. In the example illustrated in FIGS. 2-10, the coils **1141b**, **1143b**, and **1145b** are mounted on the housing **1010** through the main board **1070**. That is, the coils **1141b**, **1143b**, and **1145b** are mounted on the main board **1070**, and the main board **1070** is mounted on the housing **1010**. An example in which the main board **1070** is entirely integrally provided so that both the coils for the reflecting module **1100** and the coils for the lens module **1200** are mounted thereon is illustrated in the drawings, but the main board **1070** may be provided as two or more separate boards on which the coils for the reflecting module **1100** and the coils for the lens module **1200** are mounted, respectively.

A reinforcing plate (not illustrated) may be mounted below the main board **1070** to reinforce the main board.

In this example, when the rotation holder **1120** is rotated, a closed loop control manner of sensing and feeding back a position of the rotation holder **1120** is used.

Therefore, position sensors **1141c** and **1143c** are provided to enable a closed loop control to be performed. The position sensors **1141c** and **1143c** may be Hall sensors.

In the example illustrated in FIGS. 2-10, the position sensors **1141c** and **1143c** are disposed inside the coils **1141b** and **1143b**, respectively, and are mounted on the main board **1070** on which the coils **1141b** and **1143b** are mounted. In another example, the position sensors **1141c** and **1143c** are disposed outside the coils **1141b** and **1143b**, respectively.

The main board **1070** is provided with a gyro sensor (not illustrated) sensing shaking of the camera module **1001** by a hand-shake or other movement of the user, and is provided with a driver integrated circuit (IC) (not illustrated) providing driving signals to the coils **1141b**, **1143b**, and **1145b** based on signals from the gyro sensor.

FIGS. 11A through 11C are cross-sectional views taken along the lines XIA-XIA' through XIC-XIC' in FIG. 10 illustrating how the rotation holder of the camera module of FIG. 2 is rotated around a first axis, and FIGS. 12A through 12C are cross-sectional views taken along the lines XIIA-XIIA' through XIIC-XIIC' in FIG. 10 illustrating how the rotation holder of the camera module of FIG. 2 is rotated around a second axis.

Referring to FIGS. 11A through 11C, when the rotation holder **1120** is rotated around the first axis (the X axis), the rotation holder **1120** is rotated by rotation of the rotation plate **1130** around the first ball bearings **1131** arranged along the first axis (the X axis). In this case, the rotation holder **1120** is not moved relative to the rotation plate **1130**. In addition, referring to FIGS. 12A through 12C, when the rotation holder **1120** is rotated around the second axis (the Y axis), the rotation holder **1120** is rotated around the second

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ball bearings **1133** arranged along the second axis (the Y axis). In this case, the rotation plate **1130** is not rotated, and the rotation holder **1120** is thus moved relative to the rotation plate **1130**.

That is, when the rotation holder **1120** is rotated around the first axis (the X axis), the first ball bearings **1131** operate, and when the rotation holder **1120** is rotated around the second axis (the Y axis), the second ball bearings **1133** operate. The reason for this is that the second ball bearings **1133** aligned along the second axis (the Y axis) do not move in a state in which they are fitted into the seating grooves when the rotation holder **1120** is rotated around the first axis (the X axis), and the first ball bearings **1131** aligned along the first axis (the X axis) do not move in a state in which they are fitted into the seating grooves when the rotation holder **1120** is rotated around the second axis (the Y axis), as illustrated in the drawings.

The light of which the path is changed by the reflecting module **1100** is incident to the lens module **1200**. Therefore, optical axes of the stacked lenses provided in the lens module **1200** are aligned in the Z-axis direction, a direction in which the light is emitted from the reflecting module **1100**. In addition, the lens module **1200** includes the second driving part **1240** in order to implement the AF function and the zoom function. The lens module **1200** has a relatively low weight since it does not include components for the OIS function, and is moved in the optical axis direction in order to implement the AF function and the zoom function. Therefore, power consumption is significantly reduced.

The lens module **1200** includes a lens holder **1220** provided in the internal space of the housing **1010** and having lenses stacked therein and magnets **1241a** and **1243a** of the second driving part **1240** moving the lens holder **1220** disposed thereon.

Lenses capturing an image of a subject are stacked in the lens holder **1220**, and are mounted along an optical axis in the lens holder **1220**.

The light of which the path is changed by the reflecting module **1100** is refracted while passing through the lenses. The optical axis (the Z axis) of each of the lenses is perpendicular to the thickness direction (the Y-axis direction) of the lens module **1200**.

The lens holder **1220** is movable in the optical axis direction (the Z-axis direction) for the purpose of AF. In the example illustrated in the drawings, the lens holder **1220** is configured to be movable in a direction (including an opposite direction to the direction) in which the light of which the path is changed by the reflecting module **1100** passes through the lenses.

The second driving part **1240** generates a driving force to move the lens holder **1220** in the optical axis direction (the Z-axis direction). That is, the second driving part **1240** moves the lens holder **1220** to change a distance between the lens holder **1220** and the reflecting module **1100**.

In the example illustrated in FIGS. 2-10, the second driving part **1240** includes magnets the **1241a** and **1243a** and the coils **1241b** and **1243b** disposed to face the magnets **1241a** and **1243a**, respectively.

When power is applied to the coils **1241b** and **1243b**, the lens holder **1220** on which the magnets **1241a** and **1243a** are mounted is moved in the optical axis direction (the Z-axis direction) by an electromagnetic interaction between the magnets **1241a** and **1243a** and the coils **1241b** and **1243b**.

The magnets **1241a** and **1243a** are mounted on the lens holder **1220**. In example illustrated in FIGS. 2-10, the magnets **1241a** and **1243a** are mounted on side surfaces of the lens holder **1220**.

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The coils **1241b** and **1243b** are mounted on the housing **1010**. In the example illustrated in FIGS. 2-10, the main board **1070** is mounted on the housing **1010** in a state in which the coils **1241b** and **1243b** are mounted on the main board **1070**. A case in which both the coils for the reflecting module **1100** and the coils for the lens module **1200** are mounted on the main board **1070** is illustrated in the drawings for convenience of explanation, but the main board **1070** is not limited thereto, but may be also provided as separate boards on which the coils for the reflecting module **1100** and the coils for the lens module **1200** are mounted, respectively.

In this example, when the lens holder **1220** is moved, a closed loop control manner of sensing and feeding back a position of the lens holder **1220** is used. Therefore, a position sensor **1243c** is provided to enable a closed loop control to be performed. The position sensor **1243c** may be a Hall sensor.

In the example illustrated in FIGS. 2-10, the position sensor **1243c** is disposed inside the coil **1243b**, and is mounted on the main board **1070** on which the coil **1243b** is mounted. In another example, the position sensor **1243c** is disposed outside the coil **1243b**.

The lens holder **1220** is mounted in the housing **1010** to be movable in the optical axis direction (the Z-axis direction). In the example illustrated in the drawings, ball members **1250** are disposed between the lens holder **1220** and the housing **1010**.

The ball members **1250** serve as bearings guiding the movement of the lens holder **1220** in an AF process. In addition, the ball members **1250** also serve to maintain a spacing between the lens holder **1220** and the housing **1010**.

The ball members **1250** roll in the optical axis direction (the Z-axis direction) when a driving force in the optical axis direction (the Z-axis direction) is generated. Therefore, the ball members **1250** guide the movement of the lens holder **1220** in the optical axis direction (the Z-axis direction).

Guide grooves **1221** and **1231** accommodating the ball members **1250** are formed in at least one of surfaces of the lens holder **1220** facing the housing **1010**.

The ball members **1250** are accommodated in the guide grooves **1221** and **1231** and are fitted between the lens holder **1220** and the housing **1010**.

The guide grooves **1221** and **1231** may have a length in the optical axis direction (the Z-axis direction).

The ball members **1250** prevented from moving in the first axis direction (the X-axis direction) and the second axis direction (the Y-axis direction), and can move only in the optical axis direction (the Z-axis direction), in a state in which they are accommodated in the guide grooves **1221** and **1231**. In the example illustrated in FIGS. 2-10, the ball members **1250** can roll only in the optical axis direction (the Z-axis direction).

To this end, each of the guide grooves **1221** and **1231** is elongated in the optical axis direction (the Z-axis direction). In addition, cross sections of the guide grooves **1221** and **1231** may have various shapes such as a round shape or a polygonal shape.

The lens holder **1220** is pressed toward the housing **1010** so that the ball members **1250** are maintained in a state in which they are in contact with the lens holder **1220** and the housing **1010**.

To this end, the housing **1010** is provided with yokes **1260** facing the magnets **1241a** and **1243a** mounted on the lens holder **1220**. The yokes **1260** are formed of a magnetic material.

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An attractive force is generated between the yokes **1260** and the magnets **1241a** and **1243a**. Therefore, the lens holder **1220** can be moved in the optical axis direction (the Z-axis direction) by the driving force of the second driving part **1240** in a state in which the lens holder **1220** is in contact with the ball members **1250**.

FIG. 13 is an exploded perspective view illustrating another example of a camera module, and FIG. 14 is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. 13.

Referring to FIGS. 13 and 14, all the components of a camera module **1002** except for a reflecting module are the same as those of the camera module **1001** of FIGS. 2 through 10. Hereinafter, a configuration of the reflecting module will be described in detail below, and the same components will be denoted by the same reference numerals, and a detailed description thereof will be omitted.

The camera module **1002** includes a reflecting module **1100-2**, the lens module **1200**, and the image sensor module **1300** provided in a housing **1010-2**.

In the reflecting module **1100-2**, a rotation holder **1120-2** is supported by an inner surface of the housing **1010-2** through third ball bearings **1142** without using a separate additional component unlike the camera module **1001** of FIGS. 2 through 10. That is, the reflecting module **1100-2** does not have a component corresponding to the rotation plate **1130** of the camera module **1001** of FIGS. 2 through 10.

To this end, the housing **1010-2** has a rotation recess **1041** formed in a portion thereof facing the rotation holder **1120-2**, and the rotation holder **1120-2** has a rotation protrusion **1025** inserted into the rotation recess **1041**. The rotation recess **1041** and the rotation protrusion **1025** have a polygonal shape such as a triangular shape or a rectangular shape when viewed in the Z-axis direction to enable the rotation holder **1120-2** to perform rotation having only two degrees of freedom around predetermined axes (for example, the X axis and the Y axis). When the rotation recess **1041** and the rotation protrusion **1025** have the polygonal shape, after the rotation protrusion **1025** is inserted into the rotation recess **1041**, the rotation holder **1120-2** cannot be rotated around the Z-axis because corner portions of the rotation protrusion **1025** and the rotation recess **1041** catch on each other. The rotation recess **1041** and the rotation protrusion **1025** should not have a circular shape when viewed in the Z-axis direction because a circular shape would enable the rotation holder to be rotated around the Z-axis.

In addition, edges of the rotation recess **1041** and the rotation protrusion **1025** are provided as inclined surfaces or round surfaces so that the rotation holder **1120-2** is easily rotated around the predetermined axes (for example, the X axis and the Y axis).

In addition, opposite edges of the rotation recess **1041** are provided with third ball bearings **1142** aligned along the first axis (the X axis). In addition, the opposite edges of the rotation recess **1041** are provided with seating grooves **1042** into which the third ball bearings **1142** are inserted. The seating grooves **1042** are provided in a linear or curved shape elongated in the optical axis direction (the Z-axis direction) along the edges of the rotation recess **1041**. In addition, a cross section of the seating grooves **1042** may have various shapes such as a round shape or a polygonal shape.

In addition, the third ball bearings **1142** are provided in a state in which they are freely movable relative to, or fixed to,

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the housing 1010-2 or the rotation holder 1120-2. When the third ball bearings 1142 are provided in the state in which they are fixed to the housing 1010-2 or the rotation holder 1120-2, the third ball bearings 1142 may have a spherical shape, a hemispherical shape, or a spherical cap shape (a shape obtained by cutting a sphere by a plane to obtain a shape smaller or larger than a hemisphere. In addition, when the third ball bearings 1142 are provided in the state in which they are fixed to the housing 1010-2 or the rotation holder 1120-2, the third ball bearings 1142 may be manufactured integrally with the housing 1010-2 or the rotation holder 1120-2, or may be manufactured separately from the housing 1010-2 or the rotation holder 1120-2 and then attached to the housing 1010-2 or the rotation holder 1120-2.

Since the reflecting module 1100-2 in this example includes the third ball bearings 1142 aligned along the first axis (the X axis), the rotation holder 1120-2 may be rotated around the first axis (the X axis) or the second axis (the Y axis) perpendicular to the first axis (the X axis) in a state in which the rotation holder 1120-2 is supported by the housing 1010-2 through the third ball bearings 1142 by an attractive force between the pulling magnet 1151 and the pulling yoke 1153.

In this case, since the reflecting module 1100-2 includes only the third ball bearings 1142 aligned along the first axis (the X axis), when the rotation holder 1120-2 is rotated around the first axis (the X axis), a rotation axis of the rotation holder 1120-2 substantially corresponds to the first axis (the X axis) connecting the third ball bearings 1142 to each other, but when the rotation holder 1120-2 is rotated around the second axis (the Y axis), a rotation axis of the rotation holder 1120-2 substantially parallel with the second axis (the Y axis) is formed at a virtual point moved from the third ball bearings 1142 toward the rotation holder 1120-2 by a predetermined distance. However, a position of the rotation axis may be variously modified depending on a design shape. This will be described in more detail with reference to FIGS. 15A through 16C.

The reflecting module 1100-2 includes the reflecting member 1110-2, the rotation holder 1120-2 on which the reflecting member 1110-2 is mounted, and a first driving part 1140 generating a driving force to move the rotation holder 1120-2.

The reflecting member 1110-2 changes a path of light. For example, the reflecting member 1110-2 may be a mirror or a prism reflecting the light. The reflecting member 1110-2 is fixed to the rotation holder 1120-2.

The first driving part 1140 generates the driving force so that the rotation holder 1120-2 is rotatable around two directions. For example, the first driving part 1140 includes magnets 1141a, 1143a, and 1145a and coils 1141b, 1143b, and 1145b disposed to face the magnets 1141a, 1143a, and 1145a, and includes position sensors 1141c and 1143c sensing a position of the rotation holder 1120-2. Driving and positions of the magnets 1141a, 1143a, and 1145a, the coils 1141b, 1143b, and 1145b, and the position sensors 1141c and 1143c are the same as those described above in the camera module of FIGS. 2 through 10, and a detailed description thereof will thus be omitted.

FIGS. 15A through 15C are cross-sectional views illustrating how the rotation holder of the camera module of FIG. 13 is rotated around a first axis, and FIGS. 16A through 16C are cross-sectional views illustrating how the rotation holder of the camera module of FIG. 13 is rotated around a second axis.

Referring to FIGS. 15A through 15C, when the rotation holder 1120-2 is rotated around the first axis (the X axis), the

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rotation holder 1120-2 is rotated around the first axis (the X axis) along which the third ball bearings 1142 are arranged. In this case, since the reflecting module 1100-2 includes only the third ball bearings 1142 aligned along the first axis (the X axis), when the rotation holder 1120-2 is rotated around the first axis (the X axis), the rotation axis of the rotation holder 1120-2 is substantially an extended line connecting the third ball bearings 1142 to each other, and is substantially parallel with the first axis (the X axis).

Referring to FIGS. 16A through 16C, when the rotation holder 1120-2 is rotated around the second axis (the Y axis), a rotation axis of the rotation holder 1120-2 substantially parallel with the second axis (the Y axis) is formed at a virtual point moved from the third ball bearings 1142 toward the rotation holder 1120-2 by a predetermined distance. In this case, the rotation axis is provided in a portion spaced apart from a contact portion between the third ball bearings 1142 and the rotation holder 1120-2, and a sliding movement amount of the rotation holder 1120-2 is greater compared to a case in which the rotation holder 1120-2 is moved relative to the first axis (the X axis).

FIG. 17 is an exploded perspective view illustrating another example of a camera module, and FIG. 18 is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. 17.

Referring to FIGS. 17 and 18, all the components of a camera module 1003 except for a reflecting module are the same as those of the camera module 1001 of FIGS. 2 through 10. Hereinafter, a configuration of the reflecting module will be described in detail below, and the same components will be denoted by the same reference numerals, and a detailed description thereof will be omitted.

The camera module 1003 includes a reflecting module 1100-3, the lens module 1200, and the image sensor module 1300 provided in a housing 1010-3.

In the reflecting module 1100-3, a rotation holder 1120-3 is supported by an inner surface of the housing 1010-3 through fourth ball bearings 1143 without using a separate additional component unlike the camera module 1001 of FIGS. 2 through 10. That is, the reflecting module 1100-3 does not have a component corresponding to the rotation plate 1130 of the camera module 1001 of FIGS. 2 through 10).

To this end, the housing 1010-3 has a rotation recess 1043 formed in a portion thereof facing the rotation holder 1120-3, and the rotation holder 1120-3 is provided with a rotation protrusion 1026 inserted into the rotation recess 1043. The rotation recess 1043 and the rotation protrusion 1026 have a polygonal shape such as a triangular shape or a rectangular shape when viewed in the Z-axis direction in order for the rotation holder 1120-3 to perform only rotation having two degrees of freedom around predetermined axes (for example, the X axis and the Y axis). The reason is that when the rotation recess 1043 and the rotation protrusion 1026 have the polygonal shape, after the rotation protrusion 1026 is inserted into the rotation recess 1043, the rotation holder 1120-3 cannot be rotated around the Z axis because corner portions of the rotation protrusion 1026 and the rotation groove 1043 catch on each other.

In addition, edges of the rotation recess 1043 and the rotation protrusion 1026 may be provided as inclined surfaces or round surfaces so that the rotation holder 1120-3 is easily rotated around the predetermined axes (for example, the X axis and the Y axis).

In addition, the edges of the rotation recess 1043 may be provided with fourth ball bearings 1143 aligned along the

second axis (the Y axis) (opposite edges of the rotation recess **1043** may be provided with two fourth ball bearings aligned in the Y-axis direction). In addition, the edges of the rotation recess **1043** may be provided with seating grooves **1044** into which the fourth ball bearings **1143** are inserted. The seating grooves **1044** may be provided in a linear or curved shape to be elongate in the second axis direction (the Y-axis direction) along the edges of the rotation recess **1043**. In addition, a cross section of the seating groove **1044** may have various shapes such as a round shape or a polygonal shape.

In addition, the fourth ball bearings **1143** may be provided in a state in which they are freely movable relative to, or fixed to, the housing **1010-3** or the rotation holder **1120-3**. When the fourth ball bearings **1143** are provided in the state in which they are fixed to the housing **1010-3** or the rotation holder **1120-3**, the fourth ball bearings **1143** may be provided in a spherical or hemispherical shape (smaller or greater than a hemisphere). In addition, when the fourth ball bearings **1143** are provided in the state in which they are fixed to the housing **1010-3** or the rotation holder **1120-3**, the fourth ball bearings **1143** may be entirely manufactured integrally with the housing **1010-3** or the rotation holder **1120-3** or may be manufactured separately from the housing **1010-3** or the rotation holder **1120-3** and be then attached to the housing **1010-3** or the rotation holder **1120-3**.

Since the reflecting module **1100-3** of this example includes the fourth ball bearings **1143** aligned along the second axis (the Y axis), the rotation holder **1120-3** may be rotated around the second axis (the Y axis) or be rotated around the first axis (the X axis) perpendicular to the second axis (the Y axis), in a state in which it is supported by the housing **1010-3** through the fourth ball bearings **1143** by attractive force between the pulling magnet **1151** and the pulling yoke **1153**.

In this case, since the reflecting module **1100-3** includes only the fourth ball bearings **1143** aligned along the second axis (the Y axis), when the rotation holder **1120-3** is rotated around the second axis (the Y axis), a rotation axis of the rotation holder **1120-3** substantially corresponds to the second axis (the Y axis) connecting the fourth ball bearings **1143** to each other, but when the rotation holder **1120-3** is rotated around the first axis (the X axis), a rotation axis of the rotation holder **1120-3** substantially parallel with the first axis (the X axis) is formed at a virtual point moved from the fourth ball bearings **1143** toward the rotation holder **1120-3** by a predetermined distance. This will be described in more detail with reference to FIGS. **19A** through **20C**.

The reflecting module **1100-3** includes the reflecting member **1110-3**, the rotation holder **1120-3** on which the reflecting member **1110-3** is mounted, and a first driving part **1140** generating driving force to move the rotation holder **1120-3**.

The reflecting member **1110-3** may change a path of light. As an example, the reflecting member **1110-3** may be a mirror or a prism reflecting the light. The reflecting member **1110-3** may be fixed to the rotation holder **1120-3**.

The first driving part **1140** may generate the driving force so that the rotation holder **1120-3** is rotatable in two directions. For example, the first driving part **1140** may include a magnets **1141a**, **1143a**, and **1145a** and a coils **1141b**, **1143b**, and **1145b** disposed to face the magnets **1141a**, **1143a**, and **1145a**, and may include position sensors **1141c** and **1143c** sensing a position of the rotation holder **1120-3**. Driving and positions of the magnets **1141a**, **1143a**, and **1145a**, the coils **1141b**, **1143b**, and **1145b**, and the position sensors **1141c** and **1143c** are the same as those described

above in the camera module of FIGS. **2** through **10**, and a detailed description thereof will thus be omitted.

FIGS. **19A** through **19C** are cross-sectional views illustrating how the rotation holder of the camera module of FIG. **17** is rotated around a first axis, and FIGS. **20A** through **20C** are cross-sectional views illustrating how the rotation holder of the camera module of FIG. **17** is rotated around a second axis.

Referring to FIGS. **19A** through **19C**, when the rotation holder **1120-3** is rotated around the first axis (the X axis), the rotation axis of the rotation holder **1120-3** substantially parallel with the first axis (the X axis) is formed at the virtual point moved from the fourth ball bearings **1143** toward the rotation holder **1120-3** by the predetermined distance. In this case, the rotation axis is provided in a portion spaced apart from a contact portion between the fourth ball bearings **1143** and the rotation holder **1120-3**, and a sliding movement amount of the rotation holder **1120-3** is greater compared to a case in which the rotation holder **1120-3** is moved relative to the second axis (the Y axis).

Referring to FIGS. **20A** through **20C**, when the rotation holder **1120-3** is rotated around the second axis (the Y axis), the rotation holder **1120-3** may be rotated around the second axis (the Y axis) along which the fourth ball bearings **1143** are arranged. In this case, since the reflecting module **1100-3** includes only the fourth ball bearings **1143** aligned along the second axis (the Y axis), when the rotation holder **1120-3** is rotated around the second axis (the Y axis), the rotation axis of the rotation holder **1120-3** is substantially an extended line connecting the fourth ball bearings **1143** to each other, which is substantially parallel with the second axis (the Y axis).

FIG. **21** is an exploded perspective view illustrating another example of a camera module, and FIG. **22** is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. **21**.

Referring to FIGS. **21** and **22**, all the components of a camera module **1004** are the same as those of the camera module **1002** of FIGS. **13** and **14** except that seating grooves **1027** in which third ball bearings **1142** are seated are provided in a rotation holder **1120-2**. Hereinafter, only a configuration of the seating grooves **1027** will be briefly described below, and the same components will be denoted by the same reference numerals, and a detailed description thereof will be omitted.

The camera module **1004** includes a reflecting module **1100-2**, the lens module **1200**, and the image sensor module **1300** provided in a housing **1010-2**.

In the reflecting module **1100-2** in this example, the seating grooves **1027** in which the third ball bearings **1142** are seated are provided in the rotation holder **1120-2**, unlike in the reflecting module **1100-2** of the camera module **1002** of FIGS. **13** and **14** in which the seating grooves **1042** in which the third ball bearings **1142** are seated are provided in the edges of the rotation recess **1041**. The seating grooves **1027** are provided in a linear or curved shape to be elongate in the first axis direction (the X-axis direction) in the edges of the rotation protrusion **1025**. In addition, a cross section of the seating groove **1027** may have various shapes such as a round shape, a polygonal shape, and the like. All the components other than the seating grooves **1027** are the same as those of the camera module **1002** of FIGS. **13** and **14**, and a detailed description thereof will thus be omitted.

In addition, since only the positions at which the third ball bearings **1142** are provided are slightly different from the positions at which the third ball bearings **1142** are provided

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in the camera module **1002** of FIGS. **13** and **14** due to a structure in which the seating grooves **1027** are provided in the rotation holder **1120-2**, and a mechanism by which the rotation holder **1120-2** is rotated around the first axis (the X axis) or the second axis (the Y axis) is the same as that of the camera module **1002** of FIGS. **13** and **14**, such a mechanism is illustrated in FIGS. **15A** through **15C** and FIGS. **16A** through **16C** relating to the camera module **1002** of FIGS. **13** and **14**.

FIG. **23** is an exploded perspective view illustrating another example of a camera module, and FIG. **24** is an exploded perspective view illustrating a coupled relationship between a housing and a rotation holder of the camera module of FIG. **23**.

Referring to FIGS. **23** and **24**, all the components of a camera module **1005** are the same as those of the camera module **1003** of FIGS. **17** and **18** except that seating grooves **1028** in which fourth ball bearings **1143** are seated are provided in a rotation holder **1120-3**. Hereinafter, only a configuration of the seating grooves **1028** will be briefly described below, and the same components will be denoted by the same reference numerals, and a detailed description thereof will be omitted.

The camera module **1005** includes a reflecting module **1100-3**, the lens module **1200**, and the image sensor module **1300** provided in a housing **1010-3**.

In the reflecting module **1100-3** in this example, the seating grooves **1028** in which the fourth ball bearings **1143** are seated are provided in the rotation holder **1120-3**, unlike in the reflecting module **1100-3** of the camera module **1003** of FIGS. **17** and **18** in which the seating grooves **1044** in which the third ball bearings **1142** are seated are provided in the edges of the rotation recess **1043**. In this example, the seating grooves **1028** are provided in a linear or curved shape to be elongate in the second axis direction (the Y-axis direction) in the edges of the rotation protrusion **1026**. In addition, a cross section of the seating grooves **1028** may have various shapes such as a round shape and a polygonal shape. All the components other than the seating grooves **1028** are the same as those of the camera module **1003** of FIGS. **17** and **18**, and a detailed description thereof will thus be omitted.

In addition, since only positions in which the fourth ball bearings **1143** are provided are slightly different from the positions at which the fourth ball bearings **1143** are provided in the camera module **1003** of FIGS. **17** and **18** due to a structure in which the seating grooves **1028** are provided in the rotation holder **1120-3**, and a mechanism in which the rotation holder **1120-3** is rotated around the first axis (the X axis) or the second axis (the Y axis) is the same as that of the camera module **1003** of FIGS. **17** and **18**, such a mechanism is illustrated in FIGS. **19A** through **19C** and FIGS. **20A** through **20C** relating to the camera module **1003** of FIGS. **17** and **18**.

FIG. **25** is a perspective view illustrating another example of a portable electronic device.

Referring to FIG. **25**, a portable electronic device **2** is a portable electronic device such as a mobile communications terminal, a smartphone, or a tablet personal computer (PC) in which camera modules **500** and **1000** are mounted.

At least one of the camera modules **500** and **1000** may be the camera module **1001**, **1002**, **1003**, **1004**, or **1005** described with reference to FIGS. **2** through **24**.

That is, a portable electronic device including a dual camera module may include the camera module **1001**, **1002**, **1003**, **1004**, or **1005** as at least one of two camera modules.

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As set forth above, the reflecting module for OIS and the camera module including the same has a simple structure and a reduced size while implementing the autofocus function, the zoom function, and the OIS function. In addition, power consumption is significantly reduced.

While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

What is claimed is:

1. A reflecting module for optical image stabilization (OIS), the reflecting module comprising:

- a housing comprising an internal space;
- a rotation holder disposed in the internal space and comprising a reflecting member;
- a rotation plate rotatably disposed between the housing and the rotation holder, wherein the rotation holder is rotatably disposed on the rotation plate; and
- a driving part configured to apply a driving force to rotate the rotation holder,

wherein the rotation holder is configured to be rotatable about a first axis and a second axis substantially perpendicular to each other, and

wherein the first axis and the second axis are configured to be substantially parallel to at least one surface of the rotation plate.

2. The reflecting module of claim **1**, further comprising: a first support portion disposed on a first surface of the rotation plate and configured to form the first axis, and a second support portion disposed on a second surface of the rotation plate opposite to the first surface and configured to form the second axis.

3. The reflecting module of claim **2**, wherein the first support portion comprises a plurality of first support protrusions disposed along a direction parallel to the first axis, and

wherein the second support portion comprises a plurality of second support protrusions disposed along a direction parallel to the second axis.

4. The reflecting module of claim **2**, wherein the housing comprises a first guide groove in which at least a portion of the first support portion is accommodated, and

wherein the rotation holder comprises a second guide groove in which at least a portion of the second support portion is accommodated.

5. The reflecting module of claim **1**, wherein the driving part comprises at least one first driving magnet and at least one first driving coil disposed to face each other in the direction substantially parallel to the first axis,

wherein the at least one first driving magnet is disposed on the rotation holder, and

wherein the at least one first driving coil is disposed on the housing.

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6. The reflecting module of claim 5, wherein the driving part comprises at least one second driving magnet and at least one second driving coil disposed to face each other in the direction substantially parallel to the second axis,

wherein the at least one second driving magnet is disposed on the rotation holder,

wherein the at least one second driving coil is disposed on the housing.

7. The reflecting module of claim 6, further comprising at least one position sensor disposed inside the first driving coil and inside the second driving coil for detecting a position of the rotation holder.

8. The reflecting module of claim 6, wherein the housing comprises a plurality of openings, and

wherein the first driving coil and the second driving coil are exposed to the internal space of the housing through the plurality of openings.

9. The reflecting module of claim 1, further comprising: a first magnetic member configured to move together with the rotation holder; and

a second magnetic member disposed in the housing to face the first magnetic member,

wherein the rotation holder is supported in the housing by a magnetic force between the first magnetic member and the second magnetic member.

10. A camera module comprising:

the reflecting module of claim 1; and

a lens module comprising at least one lens configured to receive light emitted from the reflecting module.

11. The camera module of claim 10, wherein the lens module comprises:

a lens holder supporting the at least one lens and movable along a direction parallel to an optical axis of the at least one lens; and

a lens driving part configured to apply a driving force to move the lens holder,

wherein the lens driving part comprises at least one third driving magnet and at least one third driving coil disposed to face each other.

12. The camera module of claim 11, wherein the at least one third driving magnet and the at least one third driving coil are disposed to face each other in a direction perpendicular to the optical axis, and

wherein the at least one third driving magnet is disposed on the lens holder, and the at least one third driving coil is disposed on the housing.

13. The camera module of claim 12, further comprising a third magnetic member disposed in the housing to face the third driving magnet,

wherein the lens holder is supported in the housing by a magnetic force between the third driving magnet and the third magnetic member.

14. The camera module of claim 11, further comprising a guide member disposed between the lens holder and the housing to guide the movement of the lens holder.

15. A reflecting module for optical image stabilization (OIS), the reflecting module comprising:

a housing comprising an internal space;

a rotation holder rotatably disposed in the internal space of the housing and configured to support a reflecting member for changing a path of incident light; and

a rotation plate disposed between the rotation holder and the housing to rotatably support the rotation holder, wherein the rotation plate comprises:

a plurality of first support protrusions disposed on a first surface of the rotation plate along a first direction; and

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a plurality of second support protrusions disposed on a second surface of the rotation plate along a second direction substantially perpendicular to the first direction,

wherein the first surface of the rotation plate is parallel to the second surface of the rotation plate.

16. The reflecting module of claim 15, further comprising a driving part configured to apply a driving force to rotate the rotation holder,

wherein the driving part comprises:

at least one first driving magnet disposed on a side surface of the rotation holder;

at least one first driving coil disposed on the housing to face the at least one first driving magnet;

at least one second driving magnet disposed on a lower surface of the rotation holder; and

at least one second driving coil disposed on the housing to face the at least one second driving magnet.

17. A reflecting module, comprising:

a housing wall;

a rotation plate disposed on the housing wall and rotatable about a first axis relative to the housing wall;

a rotation holder disposed on the rotation plate and rotatable about a second axis relative to the rotation plate,

wherein the rotation holder is disposed to face the housing wall in a direction perpendicular to the first and second axes directions, and

wherein the rotation holder and the rotation plate are pulled toward the housing wall in the direction perpendicular to the first and second axes directions by magnetic force.

18. The reflecting module of claim 17, wherein the rotation holder comprises a reflecting member.

19. The reflecting module of claim 17, further comprising a driving part configured to apply a driving force to rotate the rotation holder about the first and second axes.

20. The reflecting module of claim 19, wherein the driving part comprises a first driving magnet and a first driving coil disposed to face each other in a direction substantially parallel to the first axis.

21. The reflecting module of claim 19, wherein the driving part comprises a second driving magnet and a second driving coil disposed to face each other in a direction substantially parallel to the second axis.

22. The reflecting module of claim 17, wherein the rotation plate comprises:

a first support portion disposed on a first surface of the rotation plate to form the first axis, and

a second support portion disposed on a second surface of the rotation plate opposite to the first surface to form the second axis.

23. The reflecting module of claim 22, wherein the first support portion comprises a plurality of first support protrusions disposed along a direction parallel to the first axis, and

wherein the second support portion comprises a plurality of second support protrusions disposed along a direction parallel to the second axis.

24. The reflecting module of claim 22, wherein the housing wall comprises a first guide groove on which the first support portion is rotatably disposed, and

wherein the rotation holder comprises a second guide groove on which the second support portion is rotatably disposed.

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25. A camera module comprising:
the reflecting module of claim **17**, wherein the rotation
holder comprises a reflecting member;
a lens holder disposed on an other housing wall, and
configured to receive light reflected from the reflecting 5
member along an optical axis; and
a guide member disposed between the lens holder and the
other housing wall,
wherein the lens holder is movable along the guide
member in a direction parallel to the optical axis. 10

26. The camera module of claim **25**, further comprising a
lens driving part configured to apply a driving force to move
the lens holder,
wherein the lens driving part comprises a third driving
magnet and a third driving coil disposed to face each 15
other in a direction substantially perpendicular to the
optical axis.

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